39. Let $\mathbf{v}$ and $\mathbf{w}$ be independent vectors in V, and let $T: V \to V'$ be a one-to-one linear transformation of V into V'. Prove that $T(\mathbf{v})$ and $T(\mathbf{w})$ are independent vectors in V'.

40. Let V and V' be vector spaces, let $B = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\}$ be a basis for V, and let $\mathbf{c}_1', \mathbf{c}_2', \ldots, \mathbf{c}_n' \in V'$. Prove that there exists a linear transformation $T: V \to V'$ such that $T(\mathbf{b}_i) = \mathbf{c}_i'$ for $i = 1, 2, \ldots, n$.

41. State and prove a generalization of Exercise 40 for any vector spaces V and V', where V has a basis B.

42. If the matrix representation of $T: \mathbb{R}^n \to \mathbb{R}^n$ relative to B,B is a diagonal matrix, describe the effect of T on the basis vectors in B.

Exercises 43 and 44 show that the set $L(V, V')$ of all linear transformations mapping a vector space V into a vector space V' is a subspace of the vector space of all functions mapping V into V'. (See summary item 5 in Section 3.1.)

43. Let T_1 and T_2 be in $L(V, V')$, and let $(T_1 + T_2): V \to V'$ be defined by

$$(T_1 + T_2)(\mathbf{v}) = T_1(\mathbf{v}) + T_2(\mathbf{v})$$

for each vector $\mathbf{v}$ in V. Prove that $T_1 + T_2$ is again a linear transformation of V into V'.

44. Let T be in $L(V, V')$, let r be any scalar in $\mathbb{R}$, and let $rT: V \to V'$ be defined by

$$(rT)(\mathbf{v}) = r(T(\mathbf{v}))$$

for each vector $\mathbf{v}$ in V. Prove that rT is again a linear transformation of V into V'.

45. If V and V' are the finite-dimensional spaces in Exercises 43 and 44 and have ordered bases B and B', respectively, describe the matrix representations of $T_1 + T_2$ in Exercise 43 and rT in Exercise 44 in terms of the matrix representations of T_1, T_2, and T relative to B,B'.

46. Prove that if $T: V \to V'$ is a linear transformation and $T(\mathbf{p}) = \mathbf{b}$, then the solution set of $T(x) = \mathbf{b}$ is $\{\mathbf{p} + \mathbf{h} \mid \mathbf{h} \in \ker(T)\}$.

47. Prove that, for any five linear transformations T_1, T_2, T_3, T_4, T_5 mapping $\mathbb{R}^2$ into $\mathbb{R}^2$, there exist scalars c_1, c_2, c_3, c_4, c_5 (not all of which are zero) such that $T = c_1T_1 + c_2T_2 + c_3T_3 + c_4T_4 + c_5T_5$ has the property that $T(\mathbf{x}) = \mathbf{0}$ for all $\mathbf{x}$ in $\mathbb{R}^2$.

48. Let $T: \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation. Prove that, if $T(T(\mathbf{x})) = T(\mathbf{x}) + T(\mathbf{x}) + 3\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$, then T is a one-to-one mapping of $\mathbb{R}^n$ into $\mathbb{R}^n$.

49. Let V and V' be vector spaces having the same finite dimension, and let $T: V \to V'$ be a linear transformation. Prove that T is one-to-one if and only if range$(T) = V'$. [HINT: Use Exercise 36 in Section 3.2.]

50. Give an example of a vector space V and a linear transformation $T: V \to V$ such that T is one-to-one but range$(T) \neq V$. [HINT: By Exercise 49, what must be true of the dimension of V?]

51. Repeat Exercise 50, but this time make range$(T) = V$ for a transformation T that is not one-to-one.

3.5 INNER-PRODUCT SPACES (Optional)

In Section 1.2, we introduced the concepts of the length of a vector and the angle between vectors in $\mathbb{R}^n$. *Length* and *angle* are defined and computed in $\mathbb{R}^n$ using the dot product of vectors. In this section, we discuss these notions for more general vector spaces. We start by recalling the properties of the dot product in $\mathbb{R}^n$, listed in Theorem 1.3, Section 1.2.

Properties of the Dot Product in $\mathbb{R}^n$

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbb{R}^n$ and let r be any scalar in $\mathbb{R}$. The following properties hold:

D1	$\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$,	**Commutative law**
D2	$\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$,	**Distributive law**
D3	$r(\mathbf{v} \cdot \mathbf{w}) = (r\mathbf{v}) \cdot \mathbf{w} = \mathbf{v} \cdot (r\mathbf{w})$,	**Homogeneity**
D4	$\mathbf{v} \cdot \mathbf{v} \geq 0$, and $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$.	**Positivity**

There are many vector spaces for which we can define useful dot products satisfying properties D1–D4. In phrasing a general definition for such vector spaces, it is customary to speak of an *inner product* rather than of a dot product, and to use the notation $\langle \mathbf{v}, \mathbf{w} \rangle$ in place of $\mathbf{v} \cdot \mathbf{w}$. One such example would be $\langle [2, 3], [4, -1] \rangle = 2(4) + 3(-1) = 5$.

DEFINITION 3.12 Inner-Product Space

An **inner product** on a vector space V is a function that associates with each ordered pair of vectors $\mathbf{v}$, $\mathbf{w}$ in V a real number, written $\langle \mathbf{v}, \mathbf{w} \rangle$, satisfying the following properties for all $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and for all scalars r:

P1	$\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$,	**Symmetry**
P2	$\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$,	**Additivity**
P3	$r\langle \mathbf{v}, \mathbf{w} \rangle = \langle r\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, r\mathbf{w} \rangle$,	**Homogeneity**
P4	$\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, and $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.	**Positivity**

An **inner-product space** is a vector space V together with an inner product on V.

EXAMPLE 1 Determine whether $\mathbb{R}^2$ is an inner-product space if, for $\mathbf{v} = [v_1, v_2]$ and $\mathbf{w} = [w_1, w_2]$, we define

$$\langle \mathbf{v}, \mathbf{w} \rangle = 2v_1w_1 + 5v_2w_2.$$

SOLUTION We check each of the four properties in Definition 3.12.

P1: Because $\langle \mathbf{v}, \mathbf{w} \rangle = 2v_1w_1 + 5v_2w_2$ and because $\langle \mathbf{w}, \mathbf{v} \rangle = 2w_1v_1 + 5w_2v_2$, the first property holds.

P2: We compute

$$\begin{aligned}
\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle &= 2u_1(v_1 + w_1) + 5u_2(v_2 + w_2), \\
\langle \mathbf{u}, \mathbf{v} \rangle &= 2u_1v_1 + 5u_2v_2, \\
\langle \mathbf{u}, \mathbf{w} \rangle &= 2u_1w_1 + 5u_2w_2.
\end{aligned}$$

The sum of the right-hand sides of the last two equations equals the right-hand side of the first equation. This establishes property P2.

P3: We compute

$$\begin{aligned} r\langle \mathbf{v}, \mathbf{w}\rangle &= r(2v_1w_1 + 5v_2w_2), \\ \langle r\mathbf{v}, \mathbf{w}\rangle &= 2(rv_1)w_1 + 5(rv_2)w_2, \\ \langle \mathbf{v}, r\mathbf{w}\rangle &= 2v_1(r_1w_1) + 5v_2(rw_2). \end{aligned}$$

Because the three right-hand expressions are equal, property P3 holds.

P4: We see that $\langle \mathbf{v}, \mathbf{v}\rangle = 2v_1^2 + 5v_2^2 \geq 0$. Because both terms of the sum are nonnegative, the sum is zero if and only if $v_1 = v_2 = 0$—that is, if and only if $\mathbf{v} = \mathbf{0}$. Therefore, we have an inner-product space. ▪

EXAMPLE 2 Determine whether $\mathbb{R}^2$ is an inner-product space when, for $\mathbf{v} = [v_1, v_2]$ and $\mathbf{w} = [w_1, w_2]$, we define

$$\langle \mathbf{v}, \mathbf{w}\rangle = 2v_1w_1 - 5v_2w_2.$$

SOLUTION A solution similar to the one for Example 1 goes along smoothly until we check property P4, which fails:

$$\langle [1, 1], [1, 1]\rangle = 2 - 5 = -3 < 0.$$

Therefore, $\mathbb{R}^2$ with this definition of $\langle\ ,\ \rangle$ is not an inner-product space. ▪

EXAMPLE 3 Determine whether the space $P_{0,1}$ of all polynomial functions with real coefficients and domain $0 \leq x \leq 1$ is an inner-product space if for p and q in $P_{0,1}$ we define

$$\langle p, q\rangle = \int_0^1 p(x)q(x)\,dx.$$

SOLUTION We check the properties for an inner product.

P1: Clearly $\langle p, q\rangle = \langle q, p\rangle$ because

$$\int_0^1 p(x)q(x)\,dx = \int_0^1 q(x)p(x)\,dx.$$

P2: For polynomial functions p, q, and h, we have

$$\begin{aligned} \langle p, q + h\rangle &= \int_0^1 p(x)(q(x) + h(x))\,dx \\ &= \int_0^1 p(x)q(x)\,dx + \int_0^1 p(x)h(x)\,dx \\ &= \langle p, q\rangle + \langle p, h\rangle. \end{aligned}$$

P3: We have

$$r\int_0^1 p(x)q(x)\,dx = \int_0^1 r(p(x))q(x)\,dx$$
$$= \int_0^1 p(x)r(q(x))\,dx,$$

so P3 holds.

P4: Because $\langle p, p\rangle = \int_0^1 p(x)^2\,dx$ and because $p(x)^2 \geq 0$ for all x, we have $\langle p, p\rangle = \int_0^1 p(x)^2\,dx \geq 0$. Now $p(x)^2$ is a continuous nonnegative polynomial function and can be zero only at a finite number of points unless $p(x)$ is the zero polynomial. It follows that

$$\langle p, p\rangle = \int_0^1 p(x)^2\,dx > 0,$$

unless $p(x)$ is the zero polynomial. This establishes P4.

Because all four properties of Definition 3.12 hold, the space $P_{0,1}$ is an inner-product space with the given inner product. ■

We mention that Example 3 is a very famous inner product, and the same definition

$$\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$$

gives an inner product on the space $C_{0,1}$ of all continuous real-valued functions with domain of the interval $0 \leq x \leq 1$. The hypothesis of continuity is essential for the demonstration of P4, as is shown in advanced calculus. Of course, there is nothing unique about the interval $0 \leq x \leq 1$. Any interval $a \leq x \leq b$ can be used in its place. The choice of interval depends on the application.

Magnitude

The condition $\langle \mathbf{v}, \mathbf{v}\rangle \geq 0$ in the definition of an inner-product space allows us to define the magnitude of a vector, just as we did in Section 1.2 using the dot product.

DEFINITION 3.13 Magnitude or Norm of a Vector

Let V be an inner-product space. The **magnitude** or **norm** of a vector $\mathbf{v}$ in V is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}$.

This definition of *norm* reduces to the usual definition of magnitude of vectors in $\mathbb{R}^n$ when the dot product is used. That is, $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$.

Recall that in $\mathbb{R}^n$ we can visualize the vector $\mathbf{v} - \mathbf{w}$ geometrically as an arrow reaching from the tip of the arrow representing $\mathbf{w}$ to the tip of the arrow representing $\mathbf{v}$, so that $\|\mathbf{v} - \mathbf{w}\|$ is the distance between the tip of $\mathbf{v}$ and the tip of $\mathbf{w}$. This leads us to define the **distance** between $\mathbf{v}$ and $\mathbf{w}$ in an inner-product space V to be $d(\mathbf{v}, \mathbf{w}) = \|\mathbf{v} - \mathbf{w}\|$.

EXAMPLE 4 In the inner-product space $P_{0,1}$ of all polynomial functions with real coefficients and domain $0 \le x \le 1$, and with inner product defined by

$$\langle p, q\rangle = \int_0^1 p(x)q(x)\,dx,$$

(a) find the magnitude of the polynomial $p(x) = x + 1$, and (b) compute the distance $d(x^2, x)$ from x^2 to x.

SOLUTION For part (a), we have

$$\begin{aligned}\|x + 1\|^2 = \langle x + 1, x + 1\rangle &= \int_0^1 (x + 1)^2\,dx\\ &= \int_0^1 (x^2 + 2x + 1)\,dx = \left(\frac{x^3}{3} + x^2 + x\right)\Big|_0^1 = \frac{7}{3}.\end{aligned}$$

Therefore, $\|x + 1\| = \sqrt{7/3}$.

For part (b), we have $d(x^2, x) = \|x^2 - x\|$. We compute

$$\begin{aligned}\|x^2 - x\|^2 = \langle x^2 - x, x^2 - x\rangle &= \int_0^1 (x^2 - x)^2\,dx\\ &= \int_0^1 (x^4 - 2x^3 + x^2)\,dx = \frac{1}{5} - \frac{1}{2} + \frac{1}{3} = \frac{1}{30}.\end{aligned}$$

Therefore, $d(x^2, x) = 1/\sqrt{30}$. ∎

The inner product used in Example 4 was not contrived just for this illustration. Another measure of the distance between the functions x and x^2 over $0 \le x \le 1$ is the maximum vertical distance between their graphs over the interval $0 \le x \le 1$, which calculus easily shows to be $\frac{1}{4}$ where $x = \frac{1}{2}$. In contrast, the inner product in this example uses an *integral* to measure the distance between the functions, not only at one point of the interval, but over the interval as a whole. Notice that the distance $1/\sqrt{30}$ we obtained is less than the maximum distance $\frac{1}{4}$ between the graphs, reflecting the fact that the graphs are less than $\frac{1}{4}$ unit apart over much of the interval. The functions are shown in Figure 3.3. The notion (used in Example 4) of distance between functions over an interval is very important in advanced mathematics, where it is used in approximating a complicated function over an interval as closely as possible by a function that is easier to handle.

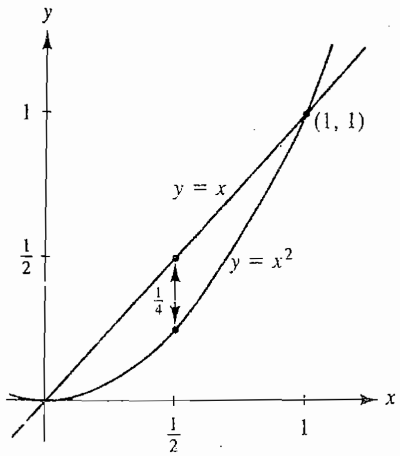

FIGURE 3.3
Graphs of x and x^2 over $0 \le x \le 1$.

It is often best, when working with the norm of a vector $\mathbf{v}$, to work with $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle$ and to introduce the radical in the final stages of the computation.

EXAMPLE 5 Let V be an inner-product space. Verify that

$$\|r\mathbf{v}\| = |r|\,\|\mathbf{v}\|$$

for any vector $\mathbf{v}$ in V and for any scalar r.

SOLUTION We have

$$\begin{aligned} \|r\mathbf{v}\|^2 &= \langle r\mathbf{v}, r\mathbf{v} \rangle && \textbf{Applying homogeneity} \\ &= r^2\langle \mathbf{v}, \mathbf{v} \rangle && \textbf{(property P3) twice} \\ &= r^2\|\mathbf{v}\|^2. \end{aligned}$$

On taking square roots, we have $\|r\mathbf{v}\| = |r|\,\|\mathbf{v}\|$. ∎

The property of norms in the preceding example can be useful in computing the magnitude of a vector and in establishing relations between magnitudes, as illustrated in the following two examples.

EXAMPLE 6 Using the standard inner product in $\mathbb{R}^n$, find the magnitude of the vector

$$\mathbf{v} = [-6, -12, 6, 18, -6].$$

SOLUTION Because $\mathbf{v} = -6[1, 2, -1, -3, 1]$, the property of norms in Example 5 tells us that

$$\|\mathbf{v}\| = |-6|\,\|[1, 2, -1, -3, 1]\| = 6\sqrt{16} = 24.$$ ∎

The Schwarz and Triangle Inequalities

In Section 1.2, we defined the angle θ between two nonzero vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$ to be

$$\theta = \arccos \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\,\|\mathbf{w}\|}.$$

The validity of this definition rested on the fact that for $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, we have

$$|\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\|\,\|\mathbf{w}\| \qquad \textbf{Schwarz inequality}$$

so that

$$-1 \leq \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\,\|\mathbf{w}\|} \leq 1.$$

The Schwarz inequality with $\mathbf{v} \cdot \mathbf{w}$ replaced by $\langle \mathbf{v}, \mathbf{w} \rangle$ holds in any inner-product space.

THEOREM 3.11 Schwarz Inequality

Let V be an inner-product space, and let $\mathbf{v}$ and $\mathbf{w}$ be vectors in V. Then $|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\|\,\|\mathbf{w}\|$.

PROOF Because the properties required for an inner-product space are modeled on those for the dot product in $\mathbb{R}^n$, we expect the proof here of the Schwarz inequality to be essentially the same as in Theorem 1.4, Section 1.2 for $\mathbb{R}^n$. This is indeed the case. Just replace every occurrence, such as $\mathbf{v} \cdot \mathbf{w}$, of a dot product in the proof of Theorem 1.4 by the corresponding inner product, such as $\langle \mathbf{v}, \mathbf{w} \rangle$. ▲

We now define the **angle between two vectors** $\mathbf{v}$ and $\mathbf{w}$ in any inner-product space to be

$$\theta = \arccos \frac{\langle \mathbf{v}, \mathbf{w} \rangle}{\|\mathbf{v}\|\,\|\mathbf{w}\|}.$$

In particular, we define $\mathbf{v}$ and $\mathbf{w}$ to be **orthogonal** (or **perpendicular**) if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$.

Recall that another important inequality in $\mathbb{R}^n$ that follows readily from the Schwarz inequality is

$$\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|. \qquad \textbf{Triangle inequality}$$

See Theorem 1.5 in Section 1.2. The triangle inequality is also valid in any inner-product space; its proof can also be obtained from the proof of Theorem 1.5 by replacing a dot product such as $\mathbf{v} \cdot \mathbf{w}$ by the corresponding inner product, $\langle \mathbf{v}, \mathbf{w} \rangle$.

SUMMARY

1. An inner-product space is a vector space V with an inner product $\langle\ ,\ \rangle$ that associates with each pair of vectors $\mathbf{v}$, $\mathbf{w}$ in V a scalar $\langle \mathbf{v}, \mathbf{w}\rangle$ that satisfies the following conditions for all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and all scalars r:

   P1 $\langle \mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{w}, \mathbf{v}\rangle$,
   P2 $\langle \mathbf{u}, \mathbf{v} + \mathbf{w}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{u}, \mathbf{w}\rangle$,
   P3 $r\langle \mathbf{v}, \mathbf{w}\rangle = \langle r\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, r\mathbf{w}\rangle$,
   P4 $\langle \mathbf{v}, \mathbf{v}\rangle \geq 0$, and $\langle \mathbf{v}, \mathbf{v}\rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.
2. $\mathbb{R}^n$ is an inner-product space using the usual dot product as inner product.
3. In an inner-product space V, the norm of a vector is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}$ and satisfies $\|r\mathbf{v}\| = |r|\,\|\mathbf{v}\|$. The distance between vectors $\mathbf{v}$ and $\mathbf{w}$ is $\|\mathbf{v} - \mathbf{w}\|$.
4. For all vectors $\mathbf{v}$ and $\mathbf{w}$ in an inner-product space, we have the following two inequalities:

   Schwarz inequality: $|\langle \mathbf{v}, \mathbf{w}\rangle| \leq \|\mathbf{v}\|\,\|\mathbf{w}\|$,
   Triangle inequality: $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$.
5. Vectors $\mathbf{v}$ and $\mathbf{w}$ in an inner-product space are orthogonal (perpendicular) if and only if $\langle \mathbf{v}, \mathbf{w}\rangle = 0$.

EXERCISES

In Exercises 1–9, determine whether or not the indicated product satisfies the conditions for an inner product in the given vector space.

1. In $\mathbb{R}^2$, let $\langle [x_1, x_2], [y_1, y_2]\rangle = x_1y_1 - x_2y_2$.
2. In $\mathbb{R}^2$, let $\langle [x_1, x_2], [y_1, y_2]\rangle = x_1x_2 + y_1y_2$.
3. In $\mathbb{R}^2$, let $\langle [x_1, x_2], [y_1, y_2]\rangle = x_1{}^2y_1 + x_2y_2$.
4. In $\mathbb{R}^2$, let $\langle [x_1, x_2], [y_1, y_2]\rangle = x_1y_2$.
5. In $\mathbb{R}^3$, let $\langle [x_1, x_2, x_3], [y_1, y_2, y_3]\rangle = x_1y_1$.
6. In $\mathbb{R}^3$, let $\langle [x_1, x_2, x_3], [y_1, y_2, y_3]\rangle = x_1 + y_1$.
7. In the vector space M_2 of all 2×2 matrices, let
$$\left\langle \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}, \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} \right\rangle = a_1b_1 + a_2b_2 + a_3b_3 + a_4b_4.$$
8. Let $C_{-1,1}$ be the vector space of all continuous functions mapping the interval $-1 \leq x \leq 1$ into $\mathbb{R}$, and let $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$.
9. Let C be as in Exercise 8, and let $\langle f, g\rangle = f(0)g(0)$.
10. Let $C_{a,b}$ be the vector space of all continuous real-valued functions with domain $a \leq x \leq b$. Prove that $\langle\ ,\ \rangle$, defined in $C_{a,b}$ by $\langle f, g\rangle = \int_a^b f(x)g(x)\,dx$, is an inner product in $C_{a,b}$.
11. Let $C_{0,1}$ be the vector space of all continuous real-valued functions with domain $0 \leq x \leq 1$. Let $\langle\ ,\ \rangle$ be defined in $C_{0,1}$ by $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$.
    a. Find $\langle (x + 1), x\rangle$.
    b. Find $\|x\|$.
    c. Find $\|x^2 - x\|$.
    d. Find $\|\sin \pi x\|$.
12. Prove that $\sin x$ and $\cos x$ are orthogonal functions in the vector space $C_{0,\pi}$ of Exercise 10, with the inner product defined there.
13. Let $\langle\ ,\ \rangle$ be defined in $C_{0,1}$ as in Exercise 11. Find a set of two independent functions in $C_{0,1}$, each of which is orthogonal to the constant function 1.
14. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner-product space, and suppose that $\|\mathbf{u}\| = 3$ and $\|\mathbf{v}\| = 5$. Find $\langle \mathbf{u} + 2\mathbf{v}, \mathbf{u} - 2\mathbf{v}\rangle$.

15. Suppose that the vectors **u** and **v** in Exercise 14 are perpendicular. Find $\langle \mathbf{u} + 2\mathbf{v}, 3\mathbf{u} + \mathbf{v} \rangle$.

16. Let V be an inner-product space. Mark each of the following True or False.

___ a. The norm of every vector in V is a positive real number.

___ b. The norm of every nonzero vector in V is a positive real number.

___ c. We have $\|r\mathbf{v}\| = r\|\mathbf{v}\|$ for every scalar r and vector **v** in V.

___ d. We have $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ for all vectors **u** and **v** in V.

___ e. Two nonzero orthogonal vectors in V are independent.

___ f. If $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ for two nonzero vectors **u** and **v** in V, then **u** and **v** are orthogonal.

___ g. An inner product can be defined on every finite-dimensional real vector space.

___ h. Let r be any real scalar. Then $\langle\ ,\ \rangle'$, defined by $\langle \mathbf{u}, \mathbf{v} \rangle' = r\langle \mathbf{u}, \mathbf{v} \rangle$ for vectors **u** and **v** in V, is also an inner product on V.

___ i. $\langle\ ,\ \rangle'$, defined in part (h), is an inner product on V if r is nonzero.

___ j. The distance between two vectors **u** and **v** in V is given by $|\langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle|$.

17. For vectors **v** and **w** in an inner-product space, prove that $\mathbf{v} - \mathbf{w}$ and $\mathbf{v} + \mathbf{w}$ are perpendicular if and only if $\|\mathbf{v}\| = \|\mathbf{w}\|$.

18. For vectors **u**, **v**, and **w** in an inner-product space and for scalars r and s, prove that, if **w** is perpendicular to both **u** and **v**, then **w** is perpendicular to $r\mathbf{u} + s\mathbf{v}$.

19. Let S be a subset of nonzero vectors in an inner-product space V, and suppose that any two different vectors in S are orthogonal. Prove that S is an independent set.

20. Let V be an inner-product space, and let S be a subset of V. Prove that

$$S^{\perp} = \{\mathbf{v} \in V \mid \mathbf{v} \text{ is orthogonal to each vector in } S\}$$

is a subspace of V.

21. Referring to Exercise 20, prove that $S \subseteq (S^{\perp})^{\perp}$.

22. Give an example of an inner-product space V for which there exists a subspace W such that $(W^{\perp})^{\perp} \neq W$.

23. (*Pythagorean theorem*) Let **u** and **v** be orthogonal vectors in an inner-product space V. Prove that $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

24. Use the triangle inequality to prove that

$$\|\mathbf{v} - \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$$

for any vectors **v** and **w** in an inner-product space V.

25. Prove that, for any vectors **v** and **w** in an inner-product space V, we have

$$\|\mathbf{v} - \mathbf{w}\| \geq \|\mathbf{v}\| - \|\mathbf{w}\|.$$

26. Prove that the vectors $\|\mathbf{v}\|\mathbf{w} + \|\mathbf{w}\|\mathbf{v}$ and $\|\mathbf{v}\|\mathbf{w} - \|\mathbf{w}\|\mathbf{v}$ in an inner-product space V are perpendicular.

27. Consider the space $C_{a,b}$ of continuous functions with domain the closed interval $a \leq x \leq b$, and let $w(x)$ be a positive continuous *weight function*, so that $w(x) > 0$ for $a \leq x \leq b$. Prove that for f and g in $C_{a,b}$, the weighted integral

$$\langle f, g \rangle = \int_a^b w(x) f(x) g(x)\, dx$$

defines an inner product on $C_{a,b}$. (Such a weight function has the effect of making the portions of the interval $a \leq x \leq b$ where $w(x)$ is large more significant than portions where $w(x)$ is smaller in computing inner products and norms.)

CHAPTER

4 DETERMINANTS

Each square matrix has associated with it a number called the *determinant* of the matrix. In Section 4.1 we introduce determinants of 2×2 and 3×3 matrices, motivated by computations of area and volume. Section 4.2 discusses determinants of $n \times n$ matrices and their properties.

Section 4.3 opens with an efficient way to compute determinants and then presents Cramer's rule as well as a formula for the inverse of an invertible square matrix in terms of determinants. Cramer's rule expresses, in terms of determinants, the solution of a square linear system having a unique solution. This method is primarily of theoretical interest because the methods presented in Chapter 1 are much more efficient for solving a square system with more than two or three equations. Because references and formulas involving Cramer's rule appear in advanced calculus and other fields, we believe that students should at least read the statement of Cramer's rule and look at an illustration.

The chapter concludes with optional Section 4.4, which discusses the significance of the determinant of the standard matrix representation of a linear transformation mapping $\mathbb{R}^n$ into $\mathbb{R}^n$. The ideas in that section form the foundation for the change-of-variable formulas for definite integrals of functions of one or more variables.

4.1 AREAS, VOLUMES, AND CROSS PRODUCTS

We introduce determinants by discussing one of their most important applications: finding areas and volumes. We will find areas and volumes of very simple boxlike regions. In calculus, one finds areas and volumes of regions having more general shapes, using formulas that involve determinants.

The Area of a Parallelogram

The parallelogram determined by two nonzero and nonparallel vectors $\mathbf{a} = [a_1, a_2]$ and $\mathbf{b} = [b_1, b_2]$ in $\mathbb{R}^2$ is shown in Figure 4.1. This parallelogram has a vertex at the origin, and we regard the arrows representing **a** and **b** as forming the two sides of the parallelogram having the origin as a common vertex.

We can find the area of this parallelogram by multiplying the length $\|\mathbf{a}\|$ of its base by the altitude h, obtaining

$$\text{Area} = \|\mathbf{a}\|\, h = \|\mathbf{a}\|\,\|\mathbf{b}\|(\sin\theta) = \|\mathbf{a}\|\,\|\mathbf{b}\|\sqrt{1-\cos^2\theta}.$$

Recall from page 24 of Section 1.2 that $\mathbf{a}\cdot\mathbf{b} = \|\mathbf{a}\|\,\|\mathbf{b}\|(\cos\theta)$. Squaring our area equation, we have

$$\begin{aligned}(\text{Area})^2 &= \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - \|\mathbf{a}\|^2\|\mathbf{b}\|^2\cos^2\theta\\ &= \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - (\mathbf{a}\cdot\mathbf{b})^2\\ &= (a_1^2 + a_2^2)(b_1^2 + b_2^2) - (a_1b_1 + a_2b_2)^2\\ &= (a_1b_2 - a_2b_1)^2.\end{aligned} \tag{1}$$

The last equality should be checked using pencil and paper. On taking square roots, we obtain

$$\text{Area} = |a_1b_2 - a_2b_1|.$$

The number within the absolute value bars is known as the **determinant** of the matrix

$$A = \begin{bmatrix} a_1 & a_2\\ b_1 & b_2\end{bmatrix}$$

and is denoted by $|A|$ or $\det(A)$, so that

$$\det(A) = \begin{vmatrix} a_1 & a_2\\ b_1 & b_2\end{vmatrix}.$$

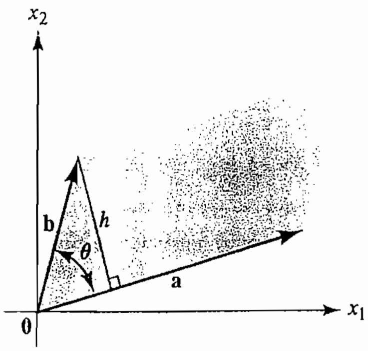

FIGURE 4.1
The parallelogram determined by **a** and **b**.

That is, if

$$A = \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix},$$

then

$$\det(A) = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1. \tag{2}$$

We can remember this formula for the determinant by taking the product of the black entries on the main diagonal of the matrix, minus the product of the colored entries on the other diagonal.

EXAMPLE 1 Find the determinant of the matrix

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}.$$

SOLUTION We have

$$\begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} = (2)(4) - (3)(1) = 5.$$

∎

EXAMPLE 2 Find the area of the parallelogram in $\mathbb{R}^2$ with vertices (1, 1), (2, 3), (2, 1), (3, 3).

SOLUTION The parallelogram is sketched in Figure 4.2. The sides having (1, 1) as common vertex can be regarded as the vectors

$$\mathbf{a} = [2, 1] - [1, 1] = [1, 0]$$

and

$$\mathbf{b} = [2, 3] - [1, 1] = [1, 2],$$

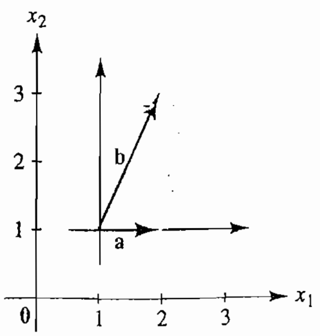

FIGURE 4.2
The parallelogram determined by $\mathbf{a} = [1, 0]$ and $\mathbf{b} = [1, 2]$.

as shown in the figure. Therefore, the area of the parallelogram is given by the determinant

$$\begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} = (1)(2) - (0)(1) = 2. \qquad \blacksquare$$

The Cross Product

Equation (2) defines a **second-order determinant**, associated with a 2×2 matrix. Another application of these second-order determinants appears when we find a vector in $\mathbb{R}^3$ that is perpendicular to each of two given independent vectors $\mathbf{b} = [b_1, b_2, b_3]$ and $\mathbf{c} = [c_1, c_2, c_3]$. Recall that the unit coordinate vectors in $\mathbb{R}^3$ are $\mathbf{i} = [1, 0, 0]$, $\mathbf{j} = [0, 1, 0]$, and $\mathbf{k} = [0, 0, 1]$. We leave as an exercise the verification that

$$\mathbf{p} = \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}\mathbf{i} - \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}\mathbf{j} + \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}\mathbf{k} \qquad (3)$$

is a vector perpendicular to both $\mathbf{b}$ and $\mathbf{c}$. (See Exercise 5.) This can be seen by computing $\mathbf{p} \cdot \mathbf{b} = \mathbf{p} \cdot \mathbf{c} = 0$. The vector $\mathbf{p}$ in formula (3) is known as the **cross product** of $\mathbf{b}$ and $\mathbf{c}$, and is denoted $\mathbf{p} = \mathbf{b} \times \mathbf{c}$.

There is a very easy way to remember formula (3) for the cross product $\mathbf{b} \times \mathbf{c}$. Form the 3×3 *symbolic matrix*

$$\begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}.$$

HISTORICAL NOTE THE NOTION OF A CROSS PRODUCT grew out of Sir William Rowan Hamilton's attempt to develop a multiplication for "triplets"—that is, vectors in $\mathbb{R}^3$. He wanted this multiplication to satisfy the associative and commutative properties as well as the distributive law. He wanted division to be always possible, except by $\mathbf{0}$. And he wanted the lengths to multiply—that is, if $(a_1, a_2, a_3)(b_1, b_2, b_3) = (c_1, c_2, c_3)$, then $\|(a_1, a_2, a_3)\|\,\|(b_1, b_2, b_3)\| = \|(c_1, c_2, c_3)\|$. After struggling with this problem for 13 years, Hamilton finally succeeded in solving it on October 16, 1843, although not in the way he had hoped. Namely, he discovered an analogous result—not for triples, but for quadruples. As he walked that day in Dublin, he wrote, he could not "resist the impulse . . . to cut with a knife on a stone of Brougham Bridge . . . the fundamental formula with the symbols i, j, k; namely, $i^2 = j^2 = k^2 = ijk = -1$." This formula symbolized his discovery of *quaternions*, elements of the form $Q = w + xi + yj + zk$ with w, x, y, z real numbers, whose multiplication obeys the laws just given, as well as the other laws Hamilton desired, except for the commutative law.

Hamilton noted the convenience of writing a quaternion Q in two parts: the scalar part w and the vector part $xi + yj + zk$. Then the product of two quaternions $\alpha = xi + yj + zk$ and $\beta = x'i + y'j + z'k$ with scalar parts 0 is given as

$$\alpha\beta = (-xx' - yy' - zz') + (yz' - zy')i + (zx' - xz')j + (xy' - yx')k.$$

The vector part of this product is our modern cross product of the "vectors" α and β, while the scalar part is the negative of the modern dot product (Section 1.2).

Although Hamilton and others pushed for the use of quaternions in physics, physicists realized by the end of the nineteenth century that the only parts of the subject necessary for their work were the two types of products of vectors. It was Josiah Willard Gibbs (1839–1903), professor of mathematical physics at Yale, who introduced our modern notation for both the dot product and the cross product and developed their properties in detail in his classes at Yale and finally in his *Vector Analysis* of 1901.

Formula (3) can be obtained from this matrix in a simple way. Multiply the vector **i** by the determinant of the 2 × 2 matrix obtained by crossing out the row and column containing **i**, as in

$$\begin{bmatrix} \cancel{\mathbf{i}} & \cancel{\mathbf{j}} & \cancel{\mathbf{k}} \\ \cancel{b_1} & b_2 & b_3 \\ \cancel{c_1} & c_2 & c_3 \end{bmatrix}.$$

Similarly, multiply $(-\mathbf{j})$ by the determinant of the matrix obtained by crossing out the row and column in which **j** appears. Finally, multiply **k** by the determinant of the matrix obtained by crossing out the row and column containing **k**, and add these multiples of **i**, **j**, and **k** to obtain formula (3).

EXAMPLE 3 Find a vector perpendicular to both [2, 1, 1] and [1, 2, 3] in $\mathbb{R}^3$.

SOLUTION We form the symbolic matrix

$$\begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

and find that

$$\begin{aligned} [2, 1, 1] \times [1, 2, 3] &= \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix}\mathbf{k} \\ &= \mathbf{i} - 5\mathbf{j} + 3\mathbf{k} = [1, -5, 3]. \end{aligned}$$

The cross product $\mathbf{p} = \mathbf{b} \times \mathbf{c}$ as defined in Eq. (3) not only is perpendicular to both **b** and **c** but points in the direction determined by the familiar *right-hand rule*: when the fingers of the right hand curve in the direction from **b** to **c**, the thumb points in the direction of $\mathbf{b} \times \mathbf{c}$. (See Figure 4.3.) We do not attempt to prove this.

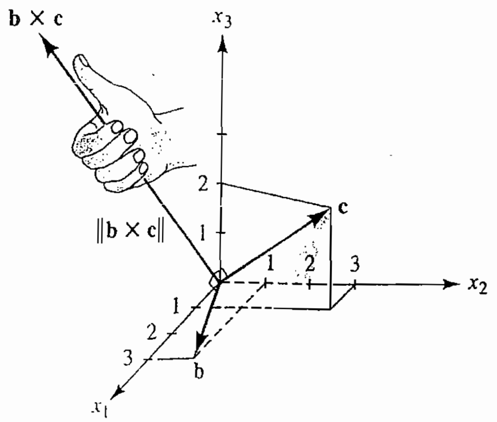

FIGURE 4.3
The area of the parallelogram determined by b and c is ||b × c||.

The magnitude of the vector $\mathbf{p} = \mathbf{b} \times \mathbf{c}$ in formula (3) is of interest as well: it is the area of the parallelogram with a vertex at the origin in $\mathbb{R}^3$ and edges at that vertex given by the vectors $\mathbf{b}$ and $\mathbf{c}$. To see this, we refer to a diagram such as the one in Figure 4.1, but with $\mathbf{a}$ replaced by $\mathbf{c}$, and we repeat the computation for area. This time, Eq. (1) takes the form

$$\begin{aligned}(\text{Area})^2 &= \|\mathbf{c}\|^2\|\mathbf{b}\|^2 - (\mathbf{c} \cdot \mathbf{b})^2 \\ &= (c_1^2 + c_2^2 + c_3^2)(b_1^2 + b_2^2 + b_3^2) - (c_1b_1 + c_2b_2 + c_3b_3)^2 \\ &= \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}^2 + \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}^2 + \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}^2.\end{aligned}$$

Again, pencil and paper are needed to check this last equality. Taking square roots, we obtain

$$\|\mathbf{b} \times \mathbf{c}\| = \text{Area of the parallelogram in } \mathbb{R}^3 \text{ determined by } \mathbf{b} \text{ and } \mathbf{c}.$$

EXAMPLE 4 Find the area of the parallelogram in $\mathbb{R}^3$ determined by the vectors $\mathbf{b} = [3, 1, 0]$ and $\mathbf{c} = [1, 3, 2]$.

SOLUTION From the symbolic matrix

$$\begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & 0 \\ 1 & 3 & 2 \end{bmatrix},$$

we find that

$$\begin{aligned}\mathbf{b} \times \mathbf{c} &= \begin{vmatrix} 1 & 0 \\ 3 & 2 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 3 & 0 \\ 1 & 2 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix}\mathbf{k} \\ &= [2, -6, 8].\end{aligned}$$

Therefore, the area of the parallelogram shown in Figure 4.3 is

$$\|\mathbf{b} \times \mathbf{c}\| = 2\sqrt{1 + 9 + 16} = 2\sqrt{26}.$$ ∎

EXAMPLE 5 Find the area of the triangle in $\mathbb{R}^3$ with vertices $(-1, 2, 0)$, $(2, 1, 3)$, and $(1, 1, -1)$.

SOLUTION We think of $(-1, 2, 0)$ as a local origin, and we take translated vectors starting there and reaching to $(2, 1, 3)$ and to $(1, 1, -1)$—namely,

$$\mathbf{a} = [2, 1, 3] - [-1, 2, 0] = [3, -1, 3]$$

and

$$\mathbf{b} = [1, 1, -1] - [-1, 2, 0] = [2, -1, -1].$$

Now $\|\mathbf{a} \times \mathbf{b}\|$ is the area of the parallelogram determined by these two vectors, and the area of the triangle is half the area of the parallelogram, as shown in Figure 4.4. We form the symbolic matrix

$$\begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 3 \\ 2 & -1 & -1 \end{bmatrix},$$

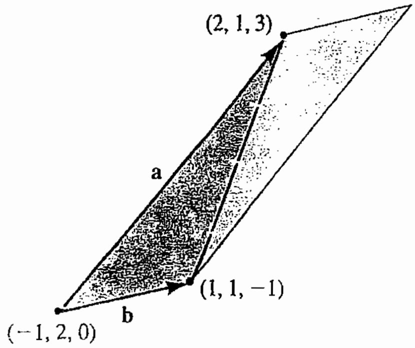

FIGURE 4.4
The triangle constitutes half the parallelogram.

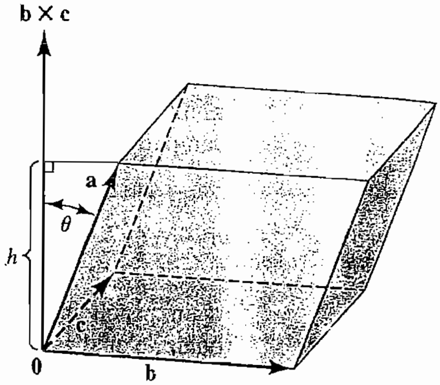

FIGURE 4.5
The box in $\mathbb{R}^3$ determined by a, b, and c.

and we find that $\mathbf{a} \times \mathbf{b} = 4\mathbf{i} + 9\mathbf{j} - \mathbf{k}$. Thus,

$$\|\mathbf{a} \times \mathbf{b}\| = \sqrt{16 + 81 + 1} = \sqrt{98} = 7\sqrt{2},$$

so the area of the triangle is $7\sqrt{2}/2$. ∎

The Volume of a Box

The cross product is useful in finding the volume of the box, or parallelepiped, determined by the three nonzero vectors $\mathbf{a} = [a_1, a_2, a_3]$, $\mathbf{b} = [b_1, b_2, b_3]$, and $\mathbf{c} = [c_1, c_2, c_3]$ in $\mathbb{R}^3$, as shown in Figure 4.5. The volume of the box can be computed by multiplying the area of the base by the altitude h.

HISTORICAL NOTE THE VOLUME INTERPRETATION of a determinant first appeared in a 1773 paper on mechanics by Joseph Louis Lagrange (1736–1813). He noted that if the points M, M', M'' have coordinates (x, y, z), (x', y', z'), (x'', y'', z''), respectively, then the tetrahedron with vertices at the origin and at those three points will have volume

$$\frac{1}{6}[z(x'y'' - y'x'') + z'(yx'' - xy'') + z''(xy' - yx')];$$

that is,

$$\frac{1}{6}\det\begin{bmatrix} x & y & z \\ x' & y' & z' \\ x'' & y'' & z'' \end{bmatrix}.$$

Lagrange was born in Turin, but spent most of his mathematical career in Berlin and in Paris. He contributed important results to such varied fields as the calculus of variation, celestial mechanics, number theory, and the theory of equations. Among his most famous works are the *Treatise on Analytical Mechanics* (1788), in which he presented the various principles of mechanics from a single point of view, and the *Theory of Analytic Functions* (1797), in which he attempted to base the differential calculus on the theory of power series.

We have just seen that the area of the base of the box is equal to $\|\mathbf{b} \times \mathbf{c}\|$, and the altitude can be found by computing

$$h = \|\mathbf{a}\|\,|\cos\theta| = \frac{\|\mathbf{b} \times \mathbf{c}\|\,\|\mathbf{a}\|\,|\cos\theta|}{\|\mathbf{b} \times \mathbf{c}\|} = \frac{|(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a}|}{\|\mathbf{b} \times \mathbf{c}\|}.$$

The absolute value is used in case $\cos\theta$ is negative. This would be the case if the direction of $\mathbf{b} \times \mathbf{c}$ were opposite to that shown in Figure 4.5. Thus,

$$\text{Volume} = (\text{Area of base})h = \|\mathbf{b} \times \mathbf{c}\|\frac{|(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a}|}{\|\mathbf{b} \times \mathbf{c}\|} = |(\mathbf{b} \times \mathbf{c}) \cdot \mathbf{a}|.$$

That is, referring to formula (3), which defines $\mathbf{b} \times \mathbf{c}$, we see that

$$\text{Volume} = |a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)|. \tag{4}$$

The number within the absolute value bars is known as a **third-order determinant**. It is the **determinant of the matrix**

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

and is denoted by

$$\det(A) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

It can be computed as

$$\det(A) = a_1\begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2\begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3\begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}. \tag{5}$$

Notice the similarity of formula (5) to our computation of the cross product $\mathbf{b} \times \mathbf{c}$ in formula (3). We simply replace $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ by a_1, a_2, and a_3, respectively.

EXAMPLE 6 Find the determinant of the matrix

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 1 & 2 \\ 1 & 2 & -3 \end{bmatrix}.$$

SOLUTION Using formula (5), we have

$$\begin{vmatrix} 2 & 1 & 3 \\ 4 & 1 & 2 \\ 1 & 2 & -3 \end{vmatrix} = 2\begin{vmatrix} 1 & 2 \\ 2 & -3 \end{vmatrix} - 1\begin{vmatrix} 4 & 2 \\ 1 & -3 \end{vmatrix} + 3\begin{vmatrix} 4 & 1 \\ 1 & 2 \end{vmatrix}$$
$$= 2(-7) - (-14) + 3(7) = 21. \quad ■$$

EXAMPLE 7 Find the volume of the box with vertex at the origin determined by the vectors $\mathbf{a} = [4, 1, 1]$, $\mathbf{b} = [2, 1, 0]$, and $\mathbf{c} = [0, 2, 3]$, and sketch the box in a figure.

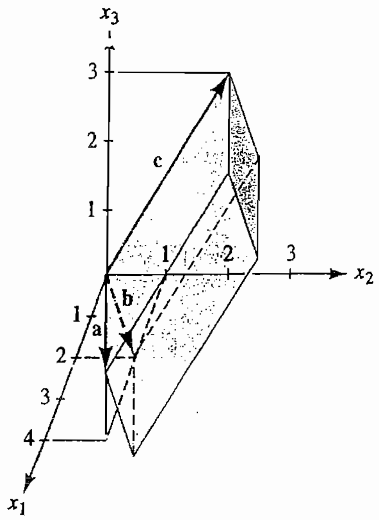

FIGURE 4.6
The box determined by a, b, and c.

SOLUTION The box is shown in Figure 4.6. Its volume is given by the absolute value of the determinant

$$\begin{vmatrix} 4 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 2 & 3 \end{vmatrix} = 4\begin{vmatrix} 1 & 0 \\ 2 & 3 \end{vmatrix} - \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix}$$
$$= 10.$$
■

The computation of $\det(A)$ in Eq. (5) is referred to as an *expansion of the determinant on the first row*. It is a particular case of a more general procedure for computing $\det(A)$, which is described in the next section.

We list the results of our work with the cross product and some of its algebraic properties in one place as a theorem. The algebraic properties can be checked through computation with the components of the vectors. Example 8 gives an illustration.

THEOREM 4.1 Properties of the Cross Product

Let **a**, **b**, and **c** be vectors in $\mathbb{R}^3$.

1. $\mathbf{b} \times \mathbf{c} = -(\mathbf{c} \times \mathbf{b})$. — Anticommutativity
2. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ is generally different from $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$. — Nonassociativity of $\times$
3. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$
   $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \times \mathbf{c}) + (\mathbf{b} \times \mathbf{c})$. — Distributive properties

4. $\mathbf{b} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{b} \times \mathbf{c}) \cdot \mathbf{c} = 0.$	Perpendicularity of $\mathbf{b} \times \mathbf{c}$ to both $\mathbf{b}$ and $\mathbf{c}$
5. $\lVert\mathbf{b} \times \mathbf{c}\rVert$ = Area of the parallelogram determined by $\mathbf{b}$ and $\mathbf{c}$.	Area property
6. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} =$ $\pm$ Volume of the box determined by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.	Volume property
7. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.$	Formula for computation of $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$

EXAMPLE 8 Show that $\mathbf{b} \times \mathbf{c} = -(\mathbf{c} \times \mathbf{b})$ for any vectors $\mathbf{b}$ and $\mathbf{c}$ in $\mathbb{R}^3$.

SOLUTION We compute

$$\mathbf{c} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \begin{vmatrix} c_2 & c_3 \\ b_2 & b_3 \end{vmatrix}\mathbf{i} - \begin{vmatrix} c_1 & c_3 \\ b_1 & b_3 \end{vmatrix}\mathbf{j} + \begin{vmatrix} c_1 & c_2 \\ b_1 & b_2 \end{vmatrix}\mathbf{k}.$$

A simple computation shows that interchanging the rows of a 2×2 matrix having a determinant d gives a matrix with determinant $-d$ (see Exercise 11). Comparison of the preceding formula for $\mathbf{c} \times \mathbf{b}$ with the formula for $\mathbf{b} \times \mathbf{c}$ in Eq. (3) then shows that $\mathbf{b} \times \mathbf{c} = -(\mathbf{c} \times \mathbf{b})$. ■

SUMMARY

1. A second-order determinant is defined by

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

   A third-order determinant is defined by Eq. (5).

2. The area of the parallelogram with vertex at the origin determined by nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^2$ is the absolute value of the determinant of the matrix having row vectors $\mathbf{a}$ and $\mathbf{b}$.

3. The cross product of vectors $\mathbf{b}$ and $\mathbf{c}$ in $\mathbb{R}^3$ can be computed by using the symbolic determinant

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

   This vector $\mathbf{b} \times \mathbf{c}$ is perpendicular to both $\mathbf{b}$ and $\mathbf{c}$.

4. The area of the parallelogram determined by nonzero vectors $\mathbf{b}$ and $\mathbf{c}$ in $\mathbb{R}^3$ is $\lVert\mathbf{b} \times \mathbf{c}\rVert$.

5. The volume of the box determined by nonzero vectors **a**, **b**, and **c** in $\mathbb{R}^3$ is the absolute value of the determinant of the matrix having row vectors **a**, **b**, and **c**. This determinant is also equal to $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.

EXERCISES

In Exercises 1–4, find the indicated determinant.

1. $\begin{vmatrix} -1 & 3 \\ 5 & 0 \end{vmatrix}$

2. $\begin{vmatrix} -1 & 0 \\ 0 & 7 \end{vmatrix}$

3. $\begin{vmatrix} 0 & -3 \\ 5 & 0 \end{vmatrix}$

4. $\begin{vmatrix} 21 & -4 \\ 10 & 7 \end{vmatrix}$

5. Show that the vector $\mathbf{p} = \mathbf{b} \times \mathbf{c}$ given in Eq. (3) is perpendicular to both **b** and **c**.

In Exercises 6–9, find the indicated determinant.

6. $\begin{vmatrix} 1 & 4 & -2 \\ 5 & 13 & 0 \\ 2 & -1 & 3 \end{vmatrix}$

7. $\begin{vmatrix} 2 & -5 & 3 \\ 1 & 3 & 4 \\ -2 & 3 & 7 \end{vmatrix}$

8. $\begin{vmatrix} 1 & -2 & 7 \\ 0 & 1 & 4 \\ 1 & 0 & 3 \end{vmatrix}$

9. $\begin{vmatrix} 2 & -1 & 1 \\ -1 & 0 & 3 \\ 2 & 1 & -4 \end{vmatrix}$

10. Show by direct computation that:

a. $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0;$

b. $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \end{vmatrix} = 0.$

11. Show by direct computation that

$$\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = -\begin{vmatrix} b_1 & b_2 \\ a_1 & a_2 \end{vmatrix}.$$

12. Show by direct computation that

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = -\begin{vmatrix} a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

In Exercises 13–18, find $\mathbf{a} \times \mathbf{b}$.

13. $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 2\mathbf{j}$

14. $\mathbf{a} = -5\mathbf{i} + \mathbf{j} + 4\mathbf{k}$, $\mathbf{b} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$

15. $\mathbf{a} = -\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$, $\mathbf{b} = 2\mathbf{i} - 4\mathbf{j} - 8\mathbf{k}$

16. $\mathbf{a} = \mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{b} = 3\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$

17. $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$, $\mathbf{b} = 4\mathbf{i} - 5\mathbf{j} + \mathbf{k}$

18. $\mathbf{a} = -2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, $\mathbf{b} = 4\mathbf{i} - 6\mathbf{j} + \mathbf{k}$

19. Mark each of the following True or False.

___ a. The determinant of a 2×2 matrix is a vector.

___ b. If two rows of a 3×3 matrix are interchanged, the sign of the determinant is changed.

___ c. The determinant of a 3×3 matrix is zero if two rows of the matrix are parallel vectors in $\mathbb{R}^3$.

___ d. In order for the determinant of a 3×3 matrix to be zero, two rows of the matrix must be parallel vectors in $\mathbb{R}^3$.

___ e. The determinant of a 3×3 matrix is zero if the points in $\mathbb{R}^3$ given by the rows of the matrix lie in a plane.

___ f. The determinant of a 3×3 matrix is zero if the points in $\mathbb{R}^3$ given by the rows of the matrix lie in a plane through the origin.

___ g. The parallelogram in $\mathbb{R}^2$ determined by nonzero vectors **a** and **b** is a square if and only if $\mathbf{a} \cdot \mathbf{b} = 0$.

___ h. The box in $\mathbb{R}^3$ determined by vectors **a**, **b**, and **c** is a cube if and only if $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c} = \mathbf{b} \cdot \mathbf{c} = 0$ and $\mathbf{a} \cdot \mathbf{a} = \mathbf{b} \cdot \mathbf{b} = \mathbf{c} \cdot \mathbf{c}$.

___ i. If the angle between vectors **a** and **b** in $\mathbb{R}^3$ is $\pi/4$, then $\|\mathbf{a} \times \mathbf{b}\| = |\mathbf{a} \cdot \mathbf{b}|$.

___ j. For any vector **a** in $\mathbb{R}^3$, we have $\|\mathbf{a} \times \mathbf{a}\| = \|\mathbf{a}\|^2$.

In Exercises 20–24, find the area of the parallelogram with vertex at the origin and with the given vectors as edges.

20. $-\mathbf{i} + 4\mathbf{j}$ and $2\mathbf{i} + 3\mathbf{j}$

21. $-5\mathbf{i} + 3\mathbf{j}$ and $\mathbf{i} + 7\mathbf{j}$

22. $\mathbf{i} + 3\mathbf{j} - 5\mathbf{k}$ and $2\mathbf{i} + 4\mathbf{j} - \mathbf{k}$

23. $2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{i} + 3\mathbf{j} - \mathbf{k}$

24. $\mathbf{i} - 4\mathbf{j} + \mathbf{k}$ and $2\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$

In Exercises 25–32, find the area of the given geometric configuration.

25. The triangle with vertices $(-1, 2)$, $(3, -1)$, and $(4, 3)$

26. The triangle with vertices $(3, -4)$, $(1, 1)$ and $(5, 7)$

27. The triangle with vertices $(2, 1, -3)$, $(3, 0, 4)$, and $(1, 0, 5)$

28. The triangle with vertices $(3, 1, -2)$, $(1, 4, 5)$, and $(2, 1, -4)$

29. The triangle in the plane $\mathbb{R}^2$ bounded by the lines $y = x$, $y = -3x + 8$, and $3y + 5x = 0$

30. The parallelogram with vertices $(1, 3)$, $(-2, 6)$, $(1, 11)$, and $(4, 8)$

31. The parallelogram with vertices $(1, 0, 1)$, $(3, 1, 4)$, $(0, 2, 9)$, and $(-2, 1, 6)$

32. The parallelogram in the plane $\mathbb{R}^2$ bounded by the lines $x - 2y = 3$, $x - 2y = 10$, $2x + 3y = -1$, and $2x + 3y = -8$

In Exercises 33–36, find $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ *and* $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$.

33. $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{b} = 4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{c} = 3\mathbf{i} + \mathbf{k}$

34. $\mathbf{a} = -\mathbf{i} + \mathbf{j} + 2\mathbf{k}$, $\mathbf{b} = \mathbf{i} + \mathbf{k}$, $\mathbf{c} = 3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$

35. $\mathbf{a} = \mathbf{i} - 3\mathbf{k}$, $\mathbf{b} = -\mathbf{i} + 4\mathbf{j}$, $\mathbf{c} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

36. $\mathbf{a} = 4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$, $\mathbf{c} = \mathbf{i} - 3\mathbf{j} + \mathbf{k}$

In Exercises 37–40, find the volume of the box having the given vectors as adjacent edges.

37. $-\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}$, $3\mathbf{i} - 2\mathbf{j} - \mathbf{k}$, $4\mathbf{i} + 2\mathbf{k}$

38. $2\mathbf{i} + \mathbf{j} - 4\mathbf{k}$, $3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{i} + 3\mathbf{j} - 8\mathbf{k}$

39. $-2\mathbf{i} + \mathbf{j}$, $3\mathbf{i} - 4\mathbf{j} + \mathbf{k}$, $\mathbf{i} - 2\mathbf{k}$

40. $3\mathbf{i} - \mathbf{j} + 4\mathbf{k}$, $\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$, $5\mathbf{i} - 3\mathbf{j} + 10\mathbf{k}$

In Exercises 41–44, find the volume of the tetrahedron having the given vertices. (Consider how the volume of a tetrahedron having three vectors from one point as edges is related to the volume of the box having the same three vectors as adjacent edges.)

41. $(-3, 0, 1)$, $(4, 2, 1)$, $(0, 1, 7)$, $(1, 1, 1)$

42. $(0, 1, 1)$, $(8, 2, -7)$, $(3, 1, 6)$, $(-4, -2, 0)$

43. $(-1, 1, 2)$, $(3, 1, 4)$, $(-1, 6, 0)$, $(2, -1, 5)$

44. $(-1, 2, 4)$, $(2, -3, 0)$, $(-4, 2, -1)$, $(0, 3, -2)$

In Exercises 45–48, use a determinant to ascertain whether the given points lie on a line in $\mathbb{R}^2$. [Hint: *What is the area of a "parallelogram" with collinear vertices?*]

45. $(0, 0)$, $(3, 5)$, $(6, 9)$

46. $(0, 0)$, $(4, 2)$, $(-6, -3)$

47. $(1, 5)$, $(3, 7)$, $(-3, 1)$

48. $(2, 3)$, $(1, -4)$, $(6, 2)$

In Exercises 49–52, use a determinant to ascertain whether the given points lie in a plane in $\mathbb{R}^3$. [Hint: *What is the "volume" of a box with coplanar vertices?*]

49. $(0, 0, 0)$, $(1, 4, 3)$, $(2, 5, 8)$, $(-1, 2, -5)$

50. $(0, 0, 0)$, $(2, 1, 1)$, $(3, -2, 1)$, $(-1, 2, 3)$

51. $(1, -1, 3)$, $(4, 2, 3)$, $(3, 1, -2)$, $(5, 5, -5)$

52. $(1, 2, 1)$, $(3, 3, 4)$, $(2, 2, 2)$, $(4, 3, 5)$

Let $\mathbf{a}$, $\mathbf{b}$, *and* $\mathbf{c}$ *be any vectors in* $\mathbb{R}^3$. *In Exercises 53–56, simplify the given expression.*

53. $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b})$

54. $(\mathbf{b} \times \mathbf{c}) - (\mathbf{c} \times \mathbf{b})$

55. $\|\mathbf{a} \times \mathbf{b}\|^2 + (\mathbf{a} \cdot \mathbf{b})^2$

56. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b})$

57. Prove property (2) of Theorem 4.1.

58. Prove property (3) of Theorem 4.1.

59. Prove property (6) of Theorem 4.1.

60. Option 7 of the routine VECTGRPH in LINTEK provides drill on the determinant of a 2×2 matrix as the area of the parallelogram determined by its row vectors, with an associated plus or minus sign. Run this option until you can regularly achieve a score of 80% or better.

MATLAB has a function **det(A)** *which gives the determinant of a matrix A. In Exercises 61–63, use the routine MATCOMP in LINTEK or MATLAB to find the volume of the box having the given vectors in* $\mathbb{R}^3$ *as adjacent edges. (We have not supplied matrix files for these problems.)*

61. $-\mathbf{i} + 7\mathbf{j} + 3\mathbf{k}$, $4\mathbf{i} + 23\mathbf{j} - 13\mathbf{k}$, $12\mathbf{i} - 17\mathbf{j} - 31\mathbf{k}$

62. $4.1\mathbf{i} - 2.3\mathbf{k}$, $5.3\mathbf{j} - 2.1\mathbf{k}$, $6.1\mathbf{i} + 5.7\mathbf{j}$

63. $2.13\mathbf{i} + 4.71\mathbf{j} - 3.62\mathbf{k}$, $5\mathbf{i} - 3.2\mathbf{j} + 6.32\mathbf{k}$, $8.3\mathbf{i} - 0.45\mathbf{j} + 1.13\mathbf{k}$

MATLAB

M1. Enter the data vectors **x** = [1 5 7] and **y** = [−3 2 4] into MATLAB. Then enter a line **crossxy** = [ ], which will compute the cross product $\mathbf{x} \times \mathbf{y}$ of vectors [x(1) x(2) x(3)] and [y(1) y(2) y(3)]. [HINT: The first component in [ ] will be **x(2)*y(3)−y(2)*x(3)**.] Be sure you use no spaces except one between the vector components. Check that the value given for **crossxy** is the correct vector $6\mathbf{i} - 25\mathbf{j} + 17\mathbf{k}$ for the data vectors entered.

M2. Use the **norm** function in MATLAB to find the area of the parallelogram in $\mathbb{R}^3$ having the vectors **x** and **y** in the preceding exercise as adjacent edges.

M3. Enter the vectors $\mathbf{x} = 4.2\mathbf{i} - 3.7\mathbf{j} + 5.6\mathbf{k}$ and $\mathbf{y} = -7.3\mathbf{i} + 4.5\mathbf{j} + 11.4\mathbf{k}$.
a. Using the up-arrow key to access your line defining **crossxy**, find $\mathbf{x} \times \mathbf{y}$.
b. Find the area of the parallelogram in $\mathbb{R}^3$ having **x** and **y** as adjacent edges.

M4. Find the area of the triangle in $\mathbb{R}^3$ having vertices (−1.2, 3.4, −6.7), (2.3, −5.2, 9.4), and (3.1, 8.3, −3.6). [HINT: Enter vectors **a**, **b**, and **c** from the origin to these points and set **x** and **y** equal to appropriate differences of them.]

NOTE: If you want to add a function **cross(x, y)** to your own personal MATLAB, do so following a procedure analogous to that described at the very end of Section 1.2 for adding the function **angl(x, y)**.

4.2 THE DETERMINANT OF A SQUARE MATRIX

The Definition

We defined a third-order determinant in terms of second-order determinants in Eq. (5) on page 245. A second-order determinant can be defined in terms of first-order determinants if we interpret the determinant of a 1×1 matrix to be its sole entry. We define an nth-order determinant in terms of determinants of order $n - 1$. In order to facilitate this, we introduce the **minor matrix** A_{ij} of an $n \times n$ matrix $A = [a_{ij}]$; it is the $(n - 1) \times (n - 1)$ matrix obtained by cross out the ith row and jth column of A. The minor matrix is the portion shown in color shading in the matrix

$$A_{ij} = \begin{array}{cc} \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix} & i\text{th row.} \\ j\text{th column} & \end{array} \qquad (1)$$

Using $|A_{ij}|$ as notation for the determinant of the minor matrix A_{ij}, we can express the determinant of a 3×3 matrix A as

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}|A_{11}| - a_{12}|A_{12}| + a_{13}|A_{13}|.$$

The numbers $a'_{11} = |A_{11}|$, $a'_{12} = -|A_{12}|$, and $a'_{13} = |A_{13}|$ are appropriately called the **cofactors** of a_{11}, a_{12}, and a_{13}. We now proceed to define the *determinant of any square matrix*, using mathematical induction. (See Appendix A for a discussion of mathematical induction.)

HISTORICAL NOTE THE FIRST APPEARANCE OF THE DETERMINANT OF A SQUARE MATRIX in Western Europe occurred in a 1683 letter from Gottfried von Leibniz (1646–1716) to the Marquis de L'Hôpital (1661–1704). Leibniz wrote a system of three equations in two unknowns with abstract "numerical" coefficients,

$$\begin{aligned} 10 + 11x + 12y &= 0 \\ 20 + 21x + 22y &= 0 \\ 30 + 31x + 32y &= 0, \end{aligned}$$

in which he noted that each coefficient number has "two characters, the first marking in which equation it occurs, the second marking which letter it belongs to." He then proceeded to eliminate first y and then x to show that the criterion for the system of equations to have a solution is that

$$10 \cdot 21 \cdot 32 + 11 \cdot 22 \cdot 30 + 12 \cdot 20 \cdot 31 = 10 \cdot 22 \cdot 31 + 11 \cdot 20 \cdot 32 + 12 \cdot 21 \cdot 30.$$

This is equivalent to the modern condition that the determinant of the matrix of coefficients must be zero.

Determinants also appeared in the contemporaneous work of the Japanese mathematician Seki Takakazu (1642–1708). Seki's manuscript of 1683 includes his detailed calculations of determinants of 2×2, 3×3, and 4×4 matrices—although his version was the negative of the version used today. Seki applied the determinant to the solving of certain types of equations, but evidently not to the solving of systems of linear equations. Seki spent most of his life as an accountant working for two feudal lords, Tokugawa Tsunashige and Tokugawa Tsunatoyo, in Kofu, a city in the prefecture of Yamanashi, west of Tokyo.

DEFINITION 4.1 Cofactors and Determinants

The **determinant** of a 1×1 matrix is its sole entry; it is a **first-order** determinant. Let $n > 1$, and assume that determinants of order less than n have been defined. Let $A = [a_{ij}]$ be an $n \times n$ matrix. The **cofactor** of a_{ij} in A is

$$a'_{ij} = (-1)^{i+j} \det(A_{ij}), \tag{2}$$

where A_{ij} given in Eq. (1) is the minor matrix of A. The **determinant** of A is

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$
$$= a_{11}a'_{11} + a_{12}a'_{12} + \cdots + a_{1n}a'_{1n}, \tag{3}$$

and is an ***n*th-order determinant.**

In addition to the notation $\det(A)$ for the determinant of A, we will sometimes use $|A|$ when determinants appear in equations, to make the equations easier to read.

EXAMPLE 1 Find the cofactor of the entry 3 in the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 2 \\ 4 & 0 & 1 & 4 \\ 1 & 0 & 2 & 1 \end{bmatrix}.$$

SOLUTION Because 3 is in the row 2, column 1 position of A, we cross out the second row and first column of A and find the cofactor of 3 to be

$$a'_{21} = (-1)^{2+1} \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 4 \\ 0 & 2 & 1 \end{vmatrix} = -\begin{vmatrix} 1 & 4 \\ 2 & 1 \end{vmatrix} - \begin{vmatrix} 0 & 1 \\ 0 & 2 \end{vmatrix}$$
$$= -(-7 - 0) = 7.$$

EXAMPLE 2 Use Eq. (3) in Definition 4.1 to find the determinant of the matrix

$$A = \begin{bmatrix} 5 & -2 & 4 & -1 \\ 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 \\ -3 & 1 & -1 & 1 \end{bmatrix}.$$

SOLUTION We have

$$\det(A) = \begin{vmatrix} 5 & -2 & 4 & -1 \\ 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 \\ -3 & 1 & -1 & 1 \end{vmatrix}$$

$$= 5(-1)^2\begin{vmatrix} 1 & 5 & 2 \\ 2 & 0 & 1 \\ 1 & -1 & 1 \end{vmatrix} + (-2)(-1)^3\begin{vmatrix} 0 & 5 & 2 \\ 1 & 0 & 1 \\ -3 & -1 & 1 \end{vmatrix}$$

$$+ 4(-1)^4\begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \\ -3 & 1 & 1 \end{vmatrix} + (-1)(-1)^5\begin{vmatrix} 0 & 1 & 5 \\ 1 & 2 & 0 \\ -3 & 1 & -1 \end{vmatrix}.$$

Computing the third-order determinants, we have

$$\begin{vmatrix} 1 & 5 & 2 \\ 2 & 0 & 1 \\ 1 & -1 & 1 \end{vmatrix} = 1\begin{vmatrix} 0 & 1 \\ -1 & 1 \end{vmatrix} - 5\begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} + 2\begin{vmatrix} 2 & 0 \\ 1 & -1 \end{vmatrix}$$
$$= 1(1) - 5(1) + 2(-2) = -8;$$

$$\begin{vmatrix} 0 & 5 & 2 \\ 1 & 0 & 1 \\ -3 & -1 & 1 \end{vmatrix} = 0\begin{vmatrix} 0 & 1 \\ -1 & 1 \end{vmatrix} - 5\begin{vmatrix} 1 & 1 \\ -3 & 1 \end{vmatrix} + 2\begin{vmatrix} 1 & 0 \\ -3 & -1 \end{vmatrix}$$
$$= 0(1) - 5(4) + 2(-1) = -22;$$

$$\begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \\ -3 & 1 & 1 \end{vmatrix} = 0\begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ -3 & 1 \end{vmatrix} + 2\begin{vmatrix} 1 & 2 \\ -3 & 1 \end{vmatrix}$$
$$= 0(1) - 1(4) + 2(7) = 10;$$

$$\begin{vmatrix} 0 & 1 & 5 \\ 1 & 2 & 0 \\ -3 & 1 & -1 \end{vmatrix} = 0\begin{vmatrix} 2 & 0 \\ 1 & -1 \end{vmatrix} - 1\begin{vmatrix} 1 & 0 \\ -3 & -1 \end{vmatrix} + 5\begin{vmatrix} 1 & 2 \\ -3 & 1 \end{vmatrix}$$
$$= 0(-2) - 1(-1) + 5(7) = 36.$$

Therefore, $\det(A) = 5(-8) + 2(-22) + 4(10) + 1(36) = -8$. ∎

The preceding example makes one thing plain:

> Computation of determinants of matrices of even moderate size using only Definition 4.1 is a tremendous chore.

According to modern astronomical theory, our solar system would be dead long before a present-day computer could find the determinant of a 50×50 matrix using just the inductive Definition 4.1. (See Exercise 37.) Section 4.3 gives an alternative, efficient method for computing determinants.

Determinants play an important role in calculus for functions of several variables. In some cases where primary values depend on some secondary values, the single number that best measures the rate at which the primary values change as the secondary values change is given by a determinant. This is closely connected with the geometric interpretation of a determinant as a *volume*. In Section 4.1, we motivated the determinant of a 2×2 matrix by using area, and we motivated the determinant of a 3×3 matrix by using volume. In Section 4.4, we show how an nth-order determinant can be interpreted as a "volume."

It is desirable to have an efficient way to compute a determinant. We will spend the remainder of this section developing properties of determinants that will enable us to find a good method for their computation. The computation of $\det(A)$ using Eq. (3) is called **expansion by minors** on the first row. Appendix B gives a proof by mathematical induction that $\det(A)$ can be obtained by using an expansion by minors *on any row or on any column*. We state this more precisely in a theorem.

THEOREM 4.2 General Expansion by Minors

Let A be an $n \times n$ matrix, and let r and s be any selections from the list of numbers $1, 2, \ldots, n$. Then

$$\det(A) = a_{r1}a'_{r1} + a_{r2}a'_{r2} + \cdots + a_{rn}a'_{rn}, \tag{4}$$

and also

$$\det(A) = a_{1s}a'_{1s} + a_{2s}a'_{2s} + \cdots + a_{ns}a'_{ns}, \tag{5}$$

where a'_{ij} is the cofactor of a_{ij} given in Definition 4.1.

Equation (4) is the **expansion of** $\det(A)$ **by minors on the rth row** of A, and Eq. (5) is the **expansion of** $\det(A)$ **by minors on the sth column** of A. Theorem 4.2 thus says that $\det(A)$ can be found by expanding by minors on any row or on any column of A.

EXAMPLE 3 Find the determinant of the matrix

$$A = \begin{bmatrix} 3 & 2 & 0 & 1 & 3 \\ -2 & 4 & 1 & 2 & 1 \\ 0 & -1 & 0 & 1 & -5 \\ -1 & 2 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}.$$

SOLUTION A recursive computation such as the one in Definition 4.1 is still the only way we have of computing $\det(A)$ at the moment, but we can expedite the computation if we expand by minors at each step on the row or column containing the most zeros. We have

$$\det(A) = 2(-1)^{5+5}\begin{vmatrix} 3 & 2 & 0 & 1 \\ -2 & 4 & 1 & 2 \\ 0 & -1 & 0 & 1 \\ -1 & 2 & 0 & -1 \end{vmatrix} \qquad \text{Expanding on row 5}$$

$$= (2)(1)(-1)^{2+3}\begin{vmatrix} 3 & 2 & 1 \\ 0 & -1 & 1 \\ -1 & 2 & -1 \end{vmatrix} \qquad \text{Expanding on column 3}$$

$$= -2\left(3(-1)^{1+1}\begin{vmatrix} -1 & 1 \\ 2 & -1 \end{vmatrix} - 1(-1)^{3+1}\begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix}\right) \qquad \text{Expanding on column 1}$$

$$= -2(3(1-2) - 1(2+1)) = -2(-3-3) = 12.$$ ∎

EXAMPLE 4 Show that the determinant of an upper- or lower-triangular square matrix is the product of its diagonal elements.

SOLUTION We work with an upper-triangular matrix, the other case being analogous. If

$$U = \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ & u_{22} & \cdots & u_{2n} \\ & & \ddots & \vdots \\ & \text{\Huge 0} & & u_{nn} \end{bmatrix}$$

is an upper-triangular matrix, then by expanding on first columns each time, we have

$$\det(U) = u_{11}\begin{vmatrix} u_{22} & u_{23} & \cdots & u_{2n} \\ 0 & u_{33} & \cdots & u_{3n} \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{vmatrix} = u_{11}u_{22}\begin{vmatrix} u_{33} & \cdots & u_{3n} \\ 0 & \cdots & u_{4n} \\ \vdots & & \vdots \\ 0 & \cdots & u_{nn} \end{vmatrix}$$

$$= \cdots = u_{11}u_{22}\cdots u_{nn}.$$ ∎

Properties of the Determinant

Using Theorem 4.2, we can establish several properties of the determinant that will be of tremendous help in its computation. Because Definition 4.1 was an inductive one, we use mathematical induction as we start to prove properties of the determinant. (Again, mathematical induction is reviewed in Appendix A.) We will always consider A to be an $n \times n$ matrix.

PROPERTY 1 The Transpose Property

For any square matrix A, we have $\det(A) = \det(A^T)$.

PROOF Verification of this property is trivial for determinants of orders 1 or 2. Let $n > 2$, and assume that the property holds for square matrices of size smaller than $n \times n$. We proceed to prove Property 1 for an $n \times n$ matrix A. We have

$$\det(A) = a_{11}|A_{11}| - a_{12}|A_{12}| + \cdots + (-1)^{n+1}a_{1n}|A_{1n}|. \quad \text{Expanding on row 1 of } A$$

Writing $B = A^T$, we have

$$\det(B) = b_{11}|B_{11}| - b_{21}|B_{21}| + \cdots + (-1)^{n+1}b_{n1}|B_{n1}|. \quad \text{Expanding on column 1 of } B$$

However, $a_{1j} = b_{j1}$ and $B_{j1} = A_{1j}^T$, because $B = A^T$. Applying our induction hypothesis to the $(n - 1)$st-order determinant $|A_{1j}|$, we have $|A_{1j}| = |B_{j1}|$. We conclude that $\det(A) = \det(B) = \det(A^T)$. ▲

This transpose property has a very useful consequence. It guarantees that any property of the determinant involving rows of a matrix is equally valid if we replace *rows* by *columns* in the statement of that property. For example, the next property has an analogue for columns.

PROPERTY 2 The Row-Interchange Property

If two different rows of a square matrix A are interchanged, the determinant of the resulting matrix is $-\det(A)$.

PROOF Again we find that the proof is trivial for the case $n = 2$. Assume that $n > 2$, and that this row-interchange property holds for matrices of size smaller than $n \times n$. Let A be an $n \times n$ matrix, and let B be the matrix obtained from A by interchanging the ith and rth rows, leaving the other rows unchanged. Because $n > 2$, we can choose a kth row for expansion by minors, where k is different from both r and i. Consider the cofactors

$$(-1)^{k+j}|A_{kj}| \quad \text{and} \quad (-1)^{k+j}|B_{kj}|.$$

These numbers must have opposite signs, by our induction hypothesis, because the minor matrices A_{kj} and B_{kj} have size $(n - 1) \times (n - 1)$, and B_{kj} can be obtained from A_{kj} by interchanging two rows. That is, $|B_{kj}| = -|A_{kj}|$. Expanding by minors on the kth row to find $\det(A)$ and $\det(B)$, we see that $\det(A) = -\det(B)$. ▲

PROPERTY 3 The Equal-Rows Property

If two rows of a square matrix A are equal, then $\det(A) = 0$.

PROOF Let B be the matrix obtained from A by interchanging the two equal rows of A. By the row-interchange property, we have $\det(B) = -\det(A)$. On the other hand, $B = A$, so $\det(A) = -\det(A)$. Therefore, $\det(A) = 0$. ▲

PROPERTY 4 The Scalar-Multiplication Property

If a single row of a square matrix A is multiplied by a scalar r, the determinant of the resulting matrix is $r \cdot \det(A)$.

PROOF Let r be any scalar, and let B be the matrix obtained from A by replacing the kth row $[a_{k1}, a_{k2}, \ldots, a_{kn}]$ of A by $[ra_{k1}, ra_{k2}, \ldots, ra_{kn}]$. Since the rows of B are equal to those of A except possibly for the kth row, it follows that the minor matrices A_{kj} and B_{kj} are equal for each j. Therefore, $a'_{kj} = b'_{kj}$, and computing $\det(B)$ by expanding by minors on the kth row, we have

$$\begin{aligned}\det(B) &= b_{k1}b'_{k1} + b_{k2}b'_{k2} + \cdots + b_{kn}b'_{kn}\\ &= r \cdot a_{k1}a'_{k1} + r \cdot a_{k2}a'_{k2} + \cdots + r \cdot a_{kn}a'_{kn}\\ &= r \cdot \det(A).\end{aligned}$$ ▲

HISTORICAL NOTE THE THEORY OF DETERMINANTS grew from the efforts of many mathematicians of the late eighteenth and early nineteenth centuries. Besides Gabriel Cramer, whose work we will discuss in the note on page 267, Etienne Bezout (1739–1783) in 1764 and Alexandre-Theophile Vandermonde (1735–1796) in 1771 gave various methods for computing determinants. In a work on integral calculus, Pierre Simon Laplace (1749–1827) had to deal with systems of linear equations. He repeated the work of Cramer, but he also stated and proved the rule that interchanging two adjacent columns of the determinant changes the sign and showed that a determinant with two equal columns will be 0.

The most complete of the early works on determinants is that of Augustin-Louis Cauchy (1789–1857) in 1812. In this work, Cauchy introduced the name *determinant* to replace several older terms, used our current double-subscript notation for a square array of numbers, defined the array of adjoints (or minors) to a given array, and showed that one can calculate the determinant by expanding on any row or column. In addition, Cauchy re-proved many of the standard theorems on determinants that had been more or less known for the past 50 years.

Cauchy was the most prolific mathematician of the nineteenth century, contributing to such areas as complex analysis, calculus, differential equations, and mechanics. In particular, he wrote the first calculus text using our modern ϵ,δ-approach to continuity. Politically he was a conservative; when the July Revolution of 1830 replaced the Bourbon king Charles X with the Orleans king Louis-Philippe, Cauchy refused to take the oath of allegiance, thereby forfeiting his chairs at the Ecole Polytechnique and the Collège de France and going into exile in Turin and Prague.

EXAMPLE 5 Find the determinant of the matrix

$$A = \begin{bmatrix} 2 & 1 & 3 & 4 & 2 \\ 6 & 2 & 1 & 4 & 1 \\ 6 & 3 & 9 & 12 & 6 \\ 2 & 1 & 3 & 4 & 1 \\ 1 & 4 & 2 & 1 & 1 \end{bmatrix}.$$

SOLUTION We note that the third row of A is three times the first row. Therefore, we have

$$\begin{aligned} \det(A) &= 3\begin{vmatrix} 2 & 1 & 3 & 4 & 2 \\ 6 & 2 & 1 & 4 & 1 \\ 2 & 1 & 3 & 4 & 2 \\ 2 & 1 & 3 & 4 & 1 \\ 1 & 4 & 2 & 1 & 1 \end{vmatrix} && \textbf{Property 4} \\ &= 3(0) = 0. && \textbf{Property 3} \end{aligned}$$

▪

The row-interchange property and the scalar-multiplication property indicate how the determinant of a matrix changes when two of the three elementary row operations are used. The next property deals with the most complicated of the elementary row operations, and lies at the heart of the efficient computation of determinants given in the next section.

PROPERTY 5 The Row-Addition Property

If the product of one row of a square matrix A by a scalar is added to a different row of A, the determinant of the resulting matrix is the same as $\det(A)$.

PROOF Let $\mathbf{a}_i = [a_{i1}, a_{i2}, \ldots, a_{in}]$ be the ith row of A. Suppose that $r\mathbf{a}_i$ is added to the kth row $\mathbf{a}_k$ of A, where r is any scalar and $k \neq i$. We obtain a matrix B whose rows are the same as the rows of A except possibly for the kth row, which is

$$\mathbf{b}_k = [ra_{i1} + a_{k1}, ra_{i2} + a_{k2}, \ldots, ra_{in} + a_{kn}].$$

Clearly the minor matrices A_{kj} and B_{kj} are equal for each j. Therefore, $a'_{kj} = b'_{kj}$, and computing $\det(B)$ by expanding by minors on the kth row, we have

$$\begin{aligned} \det(B) &= b_{k1}b'_{k1} + b_{k2}b'_{k2} + \cdots + b_{kn}b'_{kn} \\ &= (ra_{i1} + a_{k1})a'_{k1} + (ra_{i2} + a_{k2})a'_{k2} + \cdots + (ra_{in} + a_{kn})a'_{kn} \\ &= (ra_{i1}a'_{k1} + ra_{i2}a'_{k2} + \cdots + ra_{in}a'_{kn}) \\ &\quad + (a_{k1}a'_{k1} + a_{k2}a'_{k2} + \cdots + a_{kn}a'_{kn}) \\ &= r \cdot \det(C) + \det(A), \end{aligned}$$

where C is the matrix obtained from A by replacing the kth row of A with the ith row of A. Because C has two equal rows, its determinant is zero, so $\det(B) = \det(A)$. ▲

We now know how the three types of elementary row operations affect the determinant of a matrix A. In particular, if we reduce A to an echelon form H and avoid the use of row scaling, then $\det(A) = \pm\det(H)$, and $\det(H)$ is the product of its diagonal entries. (See Example 4.) We know that an echelon form of A has only nonzero entries on its main diagonal if and only if A is invertible. Thus, $\det(A) \neq 0$ if and only if A is invertible. We state this new condition for invertibility as a theorem.

THEOREM 4.3 Determinant Criterion for Invertibility

A square matrix A is invertible if and only if $\det(A) \neq 0$. Equivalently, A is singular if and only if $\det(A) = 0$.

We conclude with a multiplicative property of determinants. Section 4.4 indicates that this property has important geometric significance. Rather than labeling it "Property 6," we emphasize its increased level of importance over Properties 1 through 5 by stating it as a theorem.

THEOREM 4.4 The Multiplicative Property

If A and B are $n \times n$ matrices, then $\det(AB) = \det(A) \cdot \det(B)$.

PROOF First we note that, if A is a diagonal matrix, the result follows easily, because the product

$$\begin{bmatrix} a_{11} & & & \\ & a_{22} & & \text{\huge 0} \\ & & \ddots & \\ \text{\huge 0} & & & a_{nn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ & & \vdots & \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix}$$

has its ith row equal to a_{ii} times the ith row of B. Using the scalar-multiplication property in each of these rows, we obtain

$$\det(AB) = (a_{11}a_{22} \cdots a_{nn}) \cdot \det(B) = \det(A) \cdot \det(B).$$

To deal with the nondiagonal case, we begin by reducing the problem to the case in which the matrix A is invertible. For if A is singular, then so is AB

(see Exercise 30); so both A and AB have a zero determinant, by Theorem 4.3, and $\det(A) \cdot \det(B) = 0$, too.

If we assume that A is invertible, it can be row-reduced through row-interchange and row-addition operations to an upper-triangular matrix with nonzero entries on the diagonal. We continue such row reduction analogous to the Gauss–Jordan method but without making pivots 1, and finally we reduce A to a diagonal matrix D with nonzero diagonal entries. We can write $D = EA$, where E is the product of elementary matrices corresponding to the row interchanges and row additions used to reduce A to D. By the properties of determinants, we have $\det(A) = (-1)^r \cdot \det(D)$, where r is the number of row interchanges. The same sequence of steps will reduce the matrix AB to the matrix $E(AB) = (EA)B = DB$, so $\det(AB) = (-1)^r \cdot \det(DB)$. Therefore,

$$\det(AB) = (-1)^r \cdot \det(DB) = (-1)^r \cdot \det(D) \cdot \det(B) = \det(A) \cdot \det(B),$$

and the proof is complete. ▲

EXAMPLE 6 Find $\det(A)$ if

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}.$$

SOLUTION Because the determinant of an upper- or lower-triangular matrix is the product of the diagonal elements (see Example 4), Theorem 4.4 shows that

$$\det(A) = \begin{vmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 4 & 2 & 1 \end{vmatrix} \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{vmatrix} = (6)(2) = 12.$$ ■

EXAMPLE 7 If $\det(A) = 3$, find $\det(A^5)$ and $\det(A^{-1})$.

SOLUTION Applying Theorem 4.4 several times, we have

$$\det(A^5) = [\det(A)]^5 = 3^5 = 243.$$

From $AA^{-1} = I$, we obtain

$$1 = \det(I) = \det(AA^{-1}) = \det(A)\det(A^{-1}) = 3[\det(A^{-1})],$$

so $\det(A^{-1}) = \frac{1}{3}$. ■

Exercise 31 asks you to prove that if A is invertible, then

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

SUMMARY

1. The cofactor of an element a_{ij} in a square matrix A is $(-1)^{i+j}|A_{ij}|$, where A_{ij} is the matrix obtained from A by deleting the ith row and the jth column.
2. The *determinant* of an $n \times n$ matrix may be defined inductively by expansion by minors on the first row. The determinant can be computed by expansion by minors on any row or on any column; it is the sum of the products of the entries in that row or column by the cofactors of the entries. For large matrices, such a computation is hopelessly long.
3. The elementary row operations have the following effect on the determinant of a square matrix A.
   a. If two different rows of A are interchanged, the sign of the determinant is changed.
   b. If a single row of A is multiplied by a scalar, the determinant is multiplied by the scalar.
   c. If a multiple of one row is added to a different row, the determinant is not changed.
4. We have $\det(A) = \det(A^T)$. As a consequence, the properties just listed for elementary row operations are also true for elementary column operations.
5. If two rows or two columns of a matrix are the same, the determinant of the matrix is zero.
6. The determinant of an upper-triangular matrix or of a lower-triangular matrix is the product of the diagonal entries.
7. An $n \times n$ matrix A is invertible if and only if $\det(A) \neq 0$.
8. If A and B are $n \times n$ matrices, then $\det(AB) = \det(A) \cdot \det(B)$.

EXERCISES

In Exercises 1–10, find the determinant of the given matrix.

1. $\begin{bmatrix} 5 & 2 & 1 \\ 1 & -1 & 4 \\ 3 & 0 & 2 \end{bmatrix}$

2. $\begin{bmatrix} 1 & 0 & 6 \\ 4 & 1 & -1 \\ 5 & 0 & 1 \end{bmatrix}$

3. $\begin{bmatrix} 3 & 2 & 4 \\ 0 & 1 & 2 \\ 1 & 4 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 4 & -1 & 2 \\ 3 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$

5. $\begin{bmatrix} 0 & 1 & 4 \\ 2 & 3 & 1 \\ 1 & 4 & 1 \end{bmatrix}$

6. $\begin{bmatrix} 6 & 2 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 5 \end{bmatrix}$

7. $\begin{bmatrix} 2 & 3 & 4 & 6 \\ 2 & 0 & -9 & 6 \\ 4 & 1 & 0 & 2 \\ 0 & \mathrm{i} & -\mathrm{i} & 0 \end{bmatrix}$

8. $\begin{bmatrix} 2 & 0 & -1 & 7 \\ 6 & 1 & 0 & 4 \\ 8 & -2 & 1 & 0 \\ 4 & 1 & 0 & 2 \end{bmatrix}$

9. $\begin{bmatrix} 1 & 2 & 0 & -1 & 2 & 4 \\ 6 & 2 & 8 & 1 & -1 & 1 \\ 4 & 2 & 1 & 2 & 2 & -5 \\ 4 & 5 & 4 & 5 & 1 & 2 \\ 1 & 2 & 0 & -1 & 2 & 4 \\ 1 & 0 & 1 & 8 & 1 & 5 \end{bmatrix}$

10. $\begin{bmatrix} 1 & 0 & 1 & 2 \\ 3 & 4 & 1 & 2 \\ 6 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix}$

11. Find the cofactor of 5 for the matrix in Exercise 2.

12. Find the cofactor of 3 for the matrix in Exercise 4.

13. Find the cofactor of 7 for the matrix in Exercise 8.

14. Find the cofactor of -5 for the matrix in Exercise 9.

In Exercises 15–20, let A be a 3×3 matrix with $\det(A) = 2$.

15. Find $\det(A^2)$.

16. Find $\det(A^k)$.

17. Find $\det(3A)$.

18. Find $\det(A + A)$.

19. Find $\det(A^{-1})$.

20. Find $\det(A^T)$.

21. Mark each of the following True or False.

___ a. The determinant $\det(A)$ is defined for any matrix A.

___ b. The determinant $\det(A)$ is defined for each square matrix A.

___ c. The determinant of a square matrix is a scalar.

___ d. If a matrix A is multiplied by a scalar c, the determinant of the resulting matrix is $c \cdot \det(A)$.

___ e. If an $n \times n$ matrix A is multiplied by a scalar c, the determinant of the resulting matrix is $c^n \cdot \det(A)$.

___ f. For every square matrix A, we have $\det(AA^T) = \det(A^TA) = [\det(A)]^2$.

___ g. If two rows and also two columns of a square matrix A are interchanged, the determinant changes sign.

___ h. The determinant of an elementary matrix is nonzero.

___ i. If $\det(A) = 2$ and $\det(B) = 3$, then $\det(A + B) = 5$.

___ j. If $\det(A) = 2$ and $\det(B) = 3$, then $\det(AB) = 6$.

In Exercises 22–25, let A be a 3×3 matrix with row vectors **a**, **b**, **c** *and with determinant equal to 3. Find the determinant of the matrix having the indicated row vectors.*

22. $\mathbf{a} + \mathbf{a}, \mathbf{a} + \mathbf{b}, \mathbf{a} + \mathbf{c}$

23. $\mathbf{a}, \mathbf{b}, 2\mathbf{a} + 3\mathbf{b}$

24. $\mathbf{a}, \mathbf{b}, 2\mathbf{a} + 3\mathbf{b} + 2\mathbf{c}$

25. $\mathbf{a} + \mathbf{b}, \mathbf{b} + \mathbf{c}, \mathbf{c} + \mathbf{a}$

In Exercises 26–29, find the values of λ for which the given matrix is singular.

26. $\begin{bmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{bmatrix}$

27. $\begin{bmatrix} -\lambda & 5 \\ 2 & 3-\lambda \end{bmatrix}$

28. $\begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 1-\lambda & 4 \\ 0 & 1 & 1-\lambda \end{bmatrix}$

29. $\begin{bmatrix} 1-\lambda & 0 & 2 \\ 0 & 4-\lambda & 3 \\ 0 & 4 & -\lambda \end{bmatrix}$

30. If A and B are $n \times n$ matrices and if A is singular, prove (without using Theorem 4.4) that AB is also singular. [Hint: Assume that AB is invertible, and derive a contradiction.]

31. Prove that if A is invertible, then $\det(A^{-1}) = 1/\det(A)$.

32. If A and C are $n \times n$ matrices, with C invertible, prove that $\det(A) = \det(C^{-1}AC)$.

33. Without using the multiplicative property of determinants (Theorem 4.4), prove that $\det(AB) = \det(A) \cdot \det(B)$ for the case where B is a diagonal matrix.

34. Continuing Exercise 33, find two other types of matrices B for which it is easy to show that $\det(AB) = \det(A) \cdot \det(B)$.

35. Prove that, if three $n \times n$ matrices A, B, and C are identical except for the kth rows $\mathbf{a}_k$, $\mathbf{b}_k$, and $\mathbf{c}_k$, respectively, which are related by $\mathbf{a}_k = \mathbf{b}_k + \mathbf{c}_k$, then

$$\det(A) = \det(B) + \det(C).$$

36. Notice that

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = (a_{11}a_{22}) = (-a_{12}a_{21})$$

is a sum of signed products, where each product contains precisely one factor from each row and one factor from each column of the corresponding matrix. Prove by induction that this is true for an $n \times n$ matrix $A = [a_{ij}]$.

37. (*Application to permutation theory*) Consider an arrangement of n objects, lined up in a column. A rearrangement of the order of the objects is called a *permutation* of the objects. Every such permutation can be achieved by successively swapping the positions of pairs of the objects. For example, the first swap might be to interchange the first object with whatever one you want to be first in the new arrangement, and then continuing this procedure with the second, the third, etc. However, there are many possible sequences of swaps that will achieve a given permutation. Use the theory of determinants to prove that it is impossible to achieve the same permutation using both an even number and an odd number of swaps. [HINT: It doesn't matter what the objects actually are—think of them as being the rows of an $n \times n$ matrix.]

38. This exercise is for the reader who is skeptical of our assertion that the solar system would be dead long before a present-day computer could find the determinant of a 50×50 matrix using just Definition 4.1 with expansion by minors.

    a. Recall that $n! = n(n-1) \cdots (3)(2)(1)$. Show by induction that expansion of an $n \times n$ matrix by minors requires at least $n!$ multiplications for $n > 1$.

    b. Run the routine EBYMTIME in LINTEK and find the time required to perform $n!$ multiplications for $n = 8, 12, 16, 20, 25, 30, 40, 50, 70$, and 100.

39. Use MATLAB or the routine MATCOMP in LINTEK to check Example 2 and Exercises 5–10. Load the appropriate file of matrices if it is accessible. The determinant of a matrix A is found in MATLAB using the command **det(A)**.

4.3 COMPUTATION OF DETERMINANTS AND CRAMER'S RULE

We have seen that computation of determinants of high order is an unreasonable task if it is done directly from Definition 4.1, relying entirely on repeated expansion by minors. In the special case where a square matrix is triangular, Example 4 in Section 4.2 shows that the determinant is simply the product of the diagonal entries. We know that a matrix can be reduced to row-echelon form by means of elementary row operations, and row-echelon form for a square matrix is always triangular. The discussion leading to Theorem 4.3 in the previous section actually shows how the determinant of a matrix can be computed by a row reduction to echelon form. We rephrase part of this discussion in a box as an algorithm that a computer might follow to find a determinant.

Computation of a Determinant

The determinant of an $n \times n$ matrix A can be computed as follows:

1. Reduce A to an echelon form, using only row addition and row interchanges.
2. If any of the matrices appearing in the reduction contains a row of zeros, then $\det(A) = 0$.

3. Otherwise,

$$\det(A) = (-1)^r \cdot (\text{Product of pivots}),$$

where r is the number of row interchanges performed.

When doing a computation with pencil and paper rather than with a computer, we often use row scaling to make pivots 1, in order to ease calculations. As you study the following example, notice how the pivots accumulate as factors when the scalar-multiplication property of determinants is repeatedly used.

EXAMPLE 1 Find the determinant of the following matrix by reducing it to row-echelon form.

$$A = \begin{bmatrix} 2 & 2 & 0 & 4 \\ 3 & 3 & 2 & 2 \\ 0 & 1 & 3 & 2 \\ 2 & 0 & 2 & 1 \end{bmatrix}$$

SOLUTION We find that

$$\begin{vmatrix} 2 & 2 & 0 & 4 \\ 3 & 3 & 2 & 2 \\ 0 & 1 & 3 & 2 \\ 2 & 0 & 2 & 1 \end{vmatrix} = 2\begin{vmatrix} 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 2 \\ 0 & 1 & 3 & 2 \\ 2 & 0 & 2 & 1 \end{vmatrix} \qquad \textbf{Scalar-multiplication property}$$

$$= 2\begin{vmatrix} 1 & 1 & 0 & 2 \\ 0 & 0 & 2 & -4 \\ 0 & 1 & 3 & 2 \\ 0 & -2 & 2 & -3 \end{vmatrix} \qquad \textbf{Row-addition property twice}$$

$$= -2\begin{vmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 2 & -4 \\ 0 & -2 & 2 & -3 \end{vmatrix} \qquad \textbf{Row-interchange property}$$

$$= -2\begin{vmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 2 & -4 \\ 0 & 0 & 8 & 1 \end{vmatrix} \qquad \textbf{Row-addition property}$$

$$= (-2)(2)\begin{vmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 8 & 1 \end{vmatrix} \qquad \textbf{Scalar-multiplication property}$$

$$= (-2)(2)\begin{vmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 17 \end{vmatrix}. \quad \textbf{Row-addition property}$$

Therefore, $\det(A) = (-2)(2)(17) = -68$. ■

In our written work, we usually don't write out the shaded portion of the computation in the preceding example.

Row reduction offers an efficient way to program a computer to compute a determinant. If we are using pencil and paper, a further modification is more practical. We can use elementary row or column operations and the properties of determinants to reduce the computation to the determinant of a matrix having some row or column with a sole nonzero entry. A computer program generally modifies the matrix so that the first column has a single nonzero entry, but we can look at the matrix and choose the row or column where this can be achieved most easily. Expanding by minors on that row or column reduces the computation to a determinant of order one less, and we can continue the process until we are left with the computation of a determinant of a 2×2 matrix. Here is an illustration.

EXAMPLE 2 Find the determinant of the matrix

$$A = \begin{bmatrix} 2 & -1 & 3 & 5 \\ 2 & 0 & 1 & 0 \\ 6 & 1 & 3 & 4 \\ -7 & 3 & -2 & 8 \end{bmatrix}.$$

SOLUTION It is easiest to create zeros in the second row and then expand by minors on that row. We start by adding -2 times the third column to the first column, and we continue in this fashion:

$$\begin{vmatrix} 2 & -1 & 3 & 5 \\ 2 & 0 & 1 & 0 \\ 6 & 1 & 3 & 4 \\ -7 & 3 & -2 & 8 \end{vmatrix} = \begin{vmatrix} -4 & -1 & 3 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 3 & 4 \\ -3 & 3 & -2 & 8 \end{vmatrix} = -\begin{vmatrix} -4 & -1 & 5 \\ 0 & 1 & 4 \\ -3 & 3 & 8 \end{vmatrix}$$

$$= -\begin{vmatrix} -4 & -1 & 9 \\ 0 & 1 & 0 \\ -3 & 3 & -4 \end{vmatrix} = -\begin{vmatrix} -4 & 9 \\ -3 & -4 \end{vmatrix}$$

$$= -(16 + 27) = -43.$$ ■

Cramer's Rule

We now exhibit formulas in terms of determinants for the components in the solution vector of a square linear system $A\mathbf{x} = \mathbf{b}$, where A is an invertible matrix. The formulas are contained in the following theorem.

THEOREM 4.5 Cramer's Rule

Consider the linear system $A\mathbf{x} = \mathbf{b}$, where $A = [a_{ij}]$ is an $n \times n$ invertible matrix,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}.$$

The system has a unique solution given by

$$x_k = \frac{\det(B_k)}{\det(A)} \quad \text{for} \quad k = 1, \ldots, n, \tag{1}$$

where B_k is the matrix obtained from A by replacing the kth-column vector of A by the column vector $\mathbf{b}$.

PROOF Because A is invertible, we know that the linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution, and we let $\mathbf{x}$ be this solution. Let X_k be the matrix obtained from the $n \times n$ identity matrix by replacing its kth-column vector by the column vector $\mathbf{x}$, so that

$$X_k = \begin{bmatrix} 1 & 0 & 0 & \cdots & x_1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & x_2 & 0 & 0 & \cdots & 0 \\ & & & & \vdots & & & & \\ 0 & 0 & 0 & \cdots & x_k & 0 & 0 & \cdots & 0 \\ & & & & \vdots & & & & \\ 0 & 0 & 0 & \cdots & x_n & 0 & 0 & \cdots & 1 \end{bmatrix}.$$

Let us compute the product AX_k. If $j \neq k$, then the jth column of AX_k is the product of A and the jth column of the identity matrix, which also yields the jth column of A. If $j = k$, then the jth column of AX_k is $A\mathbf{x} = \mathbf{b}$. Thus AX_k is the matrix obtained from A by replacing the kth column of A by the column vector $\mathbf{b}$. That is, AX_k is the matrix B_k described in the statement of the theorem. From the equation $AX_k = B_k$ and the multiplicative property of determinants, we obtain

$$\det(A) \cdot \det(X_k) = \det(B_k).$$

Computing $\det(X_k)$ by expanding by minors across the kth row, we see that $\det(X_k) = x_k$, and thus $\det(A) \cdot x_k = \det(B_k)$. Because A is invertible, we know that $\det(A) \neq 0$ and so $x_k = \det(B_k)/\det(A)$ as asserted in Equation (1). ▲

EXAMPLE 3 Solve the linear system

$$\begin{aligned} 5x_1 - 2x_2 + x_3 &= 1 \\ 3x_1 + 2x_2 \phantom{{}+ x_3} &= 3 \\ x_1 + x_2 - x_3 &= 0, \end{aligned}$$

using Cramer's rule.

SOLUTION Using the notation in Theorem 4.5, we find that

$$\det(A) = \begin{vmatrix} 5 & -2 & 1 \\ 3 & 2 & 0 \\ 1 & 1 & -1 \end{vmatrix} = -15, \qquad \det(B_1) = \begin{vmatrix} 1 & -2 & 1 \\ 3 & 2 & 0 \\ 0 & 1 & -1 \end{vmatrix} = -5,$$

$$\det(B_2) = \begin{vmatrix} 5 & 1 & 1 \\ 3 & 3 & 0 \\ 1 & 0 & -1 \end{vmatrix} = -15, \qquad \det(B_3) = \begin{vmatrix} 5 & -2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 0 \end{vmatrix} = -20.$$

Hence,

$$x_1 = \frac{-5}{-15} = \frac{1}{3},$$

$$x_2 = \frac{-15}{-15} = 1,$$

$$x_3 = \frac{-20}{-15} = \frac{4}{3}.$$

■

HISTORICAL NOTE CRAMER'S RULE appeared for the first time in full generality in the *Introduction to the Analysis of Algebraic Curves* (1750) by Gabriel Cramer (1704–1752). Cramer was interested in the problem of determining the equation of a plane curve of given degree passing through a certain number of given points. For example, the general second-degree curve, whose equation is

$$A + By + Cx + Dy^2 + Exy + x^2 = 0,$$

is determined by five points. To determine A, B, C, D, and E, given the five points, Cramer substituted the coordinates of each of the points into the equation for the second-degree curve and found five linear equations for the five unknown coefficients. Cramer then referred to the appendix of the work, in which he gave his general rule: "One finds the value of each unknown by forming n fractions of which the common denominator has as many terms as there are permutations of n things." He went on to explain exactly how one calculates these terms as products of certain coefficients of the n equations, how one determines the appropriate sign for each term, and how one determines the n numerators of the fractions by replacing certain coefficients in this calculation by the constant terms of the system.

Cramer did not, however, explain why his calculations work. An explanation of the rule for the cases $n = 2$ and $n = 3$ did appear, however, in *A Treatise of Algebra* by Colin Maclaurin (1698–1746). This work was probably written in the 1730s, but was not published until 1748, after his death. In it, Maclaurin derived Cramer's rule for the two-variable case by going through the standard elimination procedure. He then derived the three-variable version by solving two pairs of equations for one unknown and equating the results, thus reducing the problem to the two-variable case. Maclaurin then described the result for the four-variable case, but said nothing about any further generalization. Interestingly, Leonhard Euler, in his *Introduction of Algebra* of 1767, does not mention Cramer's rule at all in his section on solving systems of linear equations.

The most efficient way we have presented for computing a determinant is to row-reduce a matrix to triangular form. This is also the way we solve a square linear system. If A is a 10×10 invertible matrix, solving $A\mathbf{x} = \mathbf{b}$ using Cramer's rule involves row-reducing eleven 10×10 matrices A, B_1, B_2, . . . , B_{10} to triangular form. Solving the linear system by the method of Section 1.4 requires row-reducing just one 10×11 matrix so that the first ten columns are in upper-triangular form. This illustrates the folly of using Cramer's rule to solve linear systems. However, the structure of the components of the solution vector, as given by the Cramer's rule formula $x_k = \det(B_k)/\det(A)$, is of interest in the study of advanced calculus, for example.

The Adjoint Matrix

We conclude this section by finding a formula in terms of determinants for the inverse of an invertible $n \times n$ matrix $A = [a_{ij}]$. Recall the definition of the cofactor a'_{ij} from Eq. (2) of Section 4.2. Let $A_{i\to j}$ be the matrix obtained from A by replacing the jth row of A by the ith row. That is,

$$A_{i\to j} = \begin{array}{cl} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} & \begin{array}{l} \\ \\ i\text{th row} \\ \\ j\text{th row} \\ \\ \\ \end{array} \end{array}$$

Then

$$\det(A_{i\to j}) = \begin{cases} \det(A) & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

If we expand $\det(A_{i\to j})$ by minors on the jth row, we have

$$\det(A_{i\to j}) = \sum_{s=1}^{n} a_{is}a'_{js},$$

and we obtain the important relation

$$\sum_{s=1}^{n} a_{is}a'_{js} = \begin{cases} \det(A) & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \qquad (2)$$

The term on the left-hand side in Eq. (2) is the entry in the ith row and jth column in the product $A(A')^T$, where $A' = [a'_{ij}]$ is the matrix whose entries are the cofactors of the entries of A. Thus Eq. (2) can be written in matrix form as

$$A(A')^T = (\det(A))I,$$